# Super-Resolution Model Evaluation Report

Models: SRCNN, DSCMS, PRUSR, SoFoNO, SRNO

2026-08-20

Test set: HYCOM Gulf of Mexico ocean-current fields (u, v), years 2022-2024.
Degradation: downsample_factor=1 (no-op), Gaussian frequency low-pass filter
(sigma=10) -- matches the exact LR-construction used in train.py.

Metrics (per channel u, v, averaged over the full test set):
  MSE, SSIM, PSNR, LPIPS, FSIM, EPI, TKE
Metrics (channel-combined, since kinetic energy mixes u and v):
  KE  - KineticEnergySpectrumLoss (log-spectral MSE)
  WKE - WeightedKineticEnergySpectrumLoss (high-wavenumber-weighted)

Qualitative plots use one fixed test-set sample (index 0), identical across
every model/strategy shown, so figures are directly comparable.

Bold values mark the best strategy per metric, within each model's own table.

Strategies evaluated:
- mse: MSE only
- mse_lpips: 0.6 MSE + 0.4 LPIPS
- mse_lpips_ke: 0.6 MSE + 0.2 LPIPS + 0.2 KE
- mse_wke: 0.6 MSE + 0.4 Weighted KE

Lower is better: ke, lpips, mse, tke, wke
Higher is better: epi, fsim, psnr, ssim

# SRCNN -- Summary metrics (bold = best per column)

## u-velocity metrics

| Strategy | MSE | SSIM | PSNR | LPIPS | FSIM | EPI | TKE |
|---|---|---|---|---|---|---|---|
| MSE only | 0.001156 | 0.9517 | 37.75 | 0.2305 | 0.6355 | **0.0352** | 0.02228 |
| 0.6 MSE + 0.4 LPIPS | **0.0005498** | **0.9756** | **40.98** | **0.07401** | **0.7416** | 0.03022 | **0.009158** |
| 0.6 MSE + 0.2 LPIPS + 0.2 KE | 0.0006288 | 0.9728 | 40.44 | 0.08062 | 0.7257 | 0.0287 | 0.01135 |
| 0.6 MSE + 0.4 Weighted KE | 0.003569 | 0.9309 | 33.25 | 0.2612 | 0.5509 | 0.02049 | 0.05662 |

## v-velocity metrics

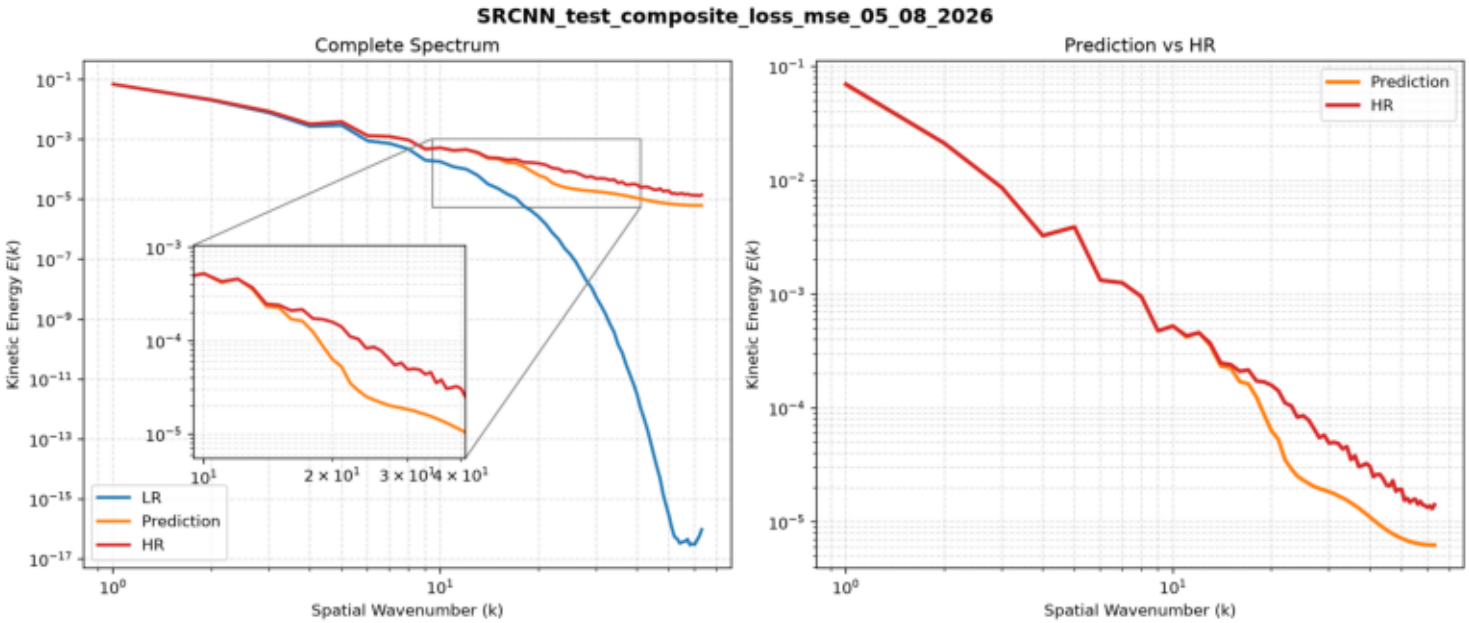
| Strategy | MSE | SSIM | PSNR | LPIPS | FSIM | EPI | TKE |
|---|---|---|---|---|---|---|---|
| MSE only | 0.001184 | 0.9557 | 38.5 | 0.228 | 0.7009 | **0.03094** | 0.02392 |
| 0.6 MSE + 0.4 LPIPS | **0.0005911** | **0.9777** | **41.49** | **0.07062** | **0.7911** | 0.02668 | **0.009994** |
| 0.6 MSE + 0.2 LPIPS + 0.2 KE | 0.0006719 | 0.9753 | 40.99 | 0.07683 | 0.7768 | 0.02538 | 0.01224 |
| 0.6 MSE + 0.4 Weighted KE | 0.002677 | 0.9394 | 34.99 | 0.2531 | 0.6517 | 0.0254 | 0.04288 |

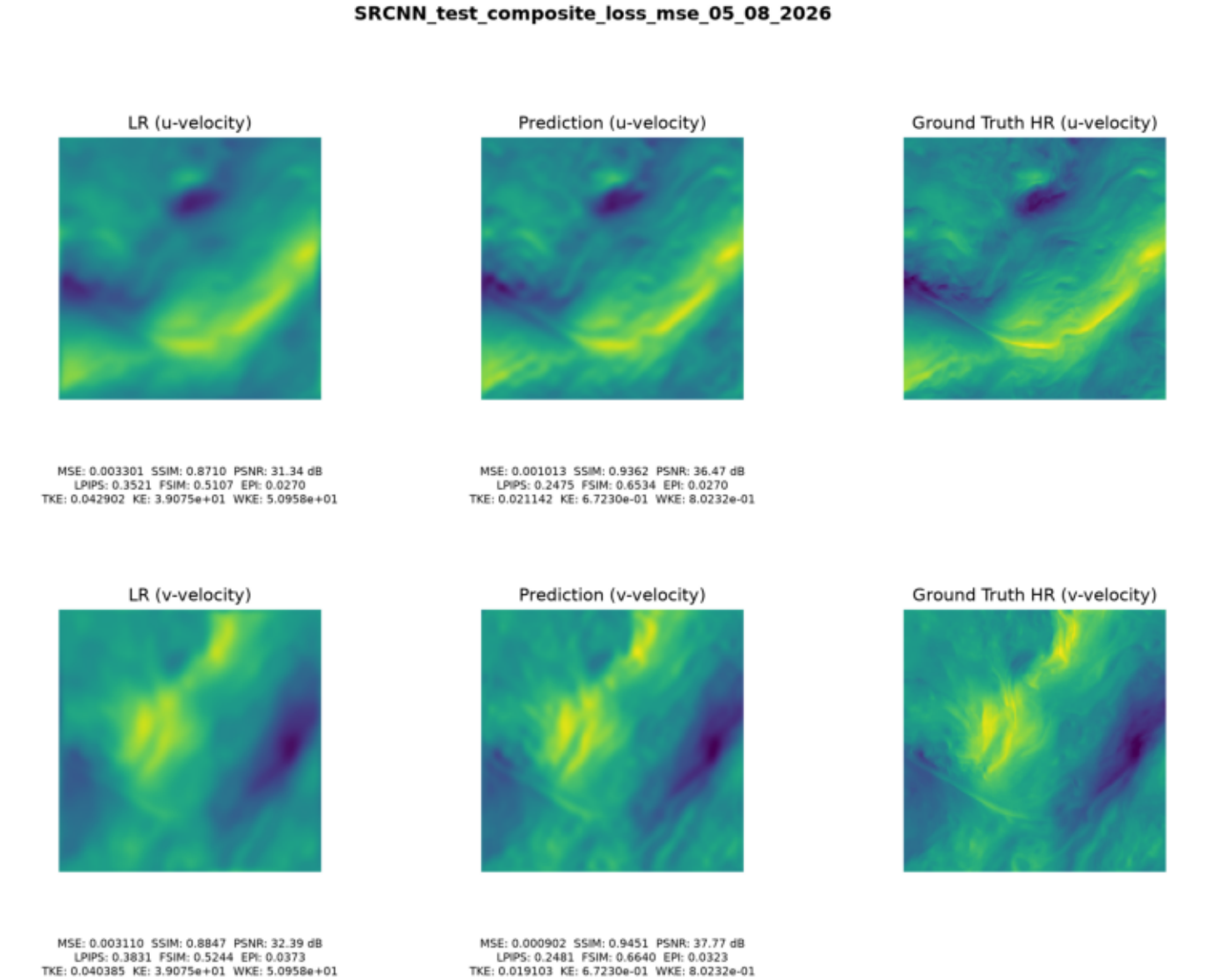
## Combined metrics (KE / WKE mix u and v, not split by channel)

| Strategy | KE | WKE |
|---|---|---|
| MSE only | 0.436 | 0.5251 |
| 0.6 MSE + 0.4 LPIPS | 0.04043 | 0.05075 |
| 0.6 MSE + 0.2 LPIPS + 0.2 KE | **0.005288** | **0.007215** |
| 0.6 MSE + 0.4 Weighted KE | 0.0134 | 0.01579 |

# SRCNN_test_composite_loss_mse_05_08_2026

MSE only

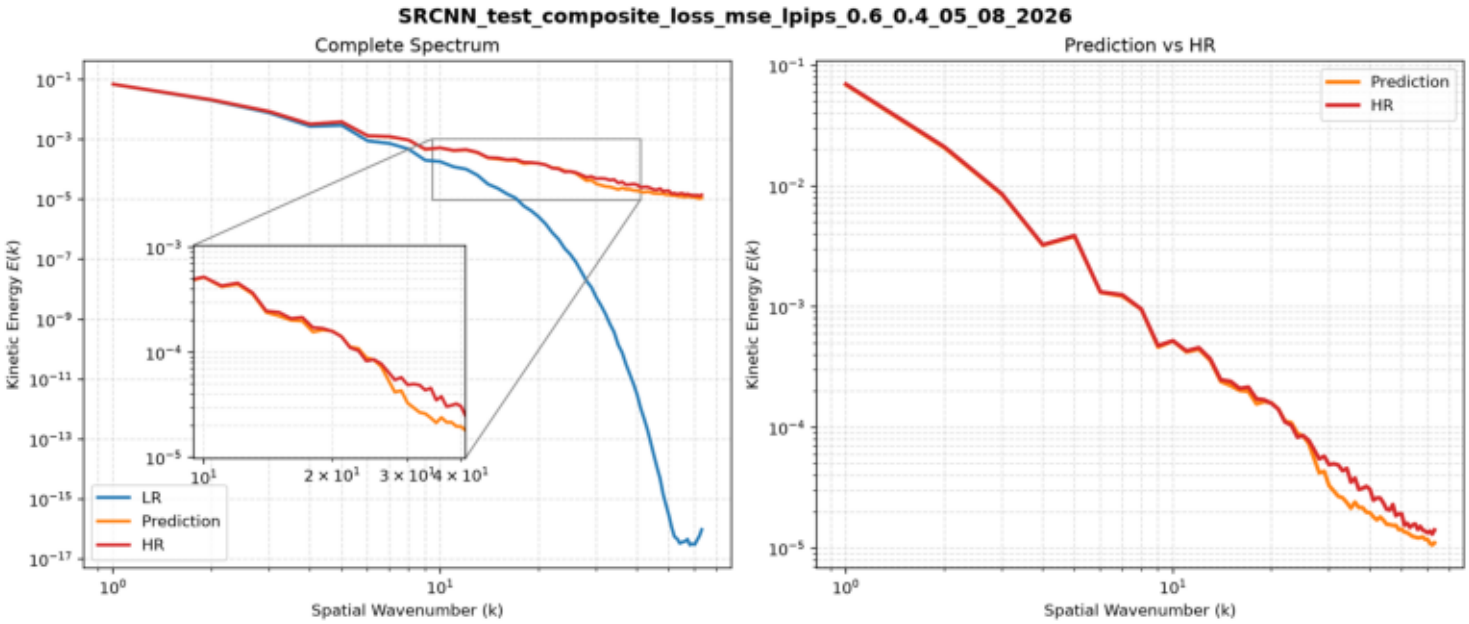


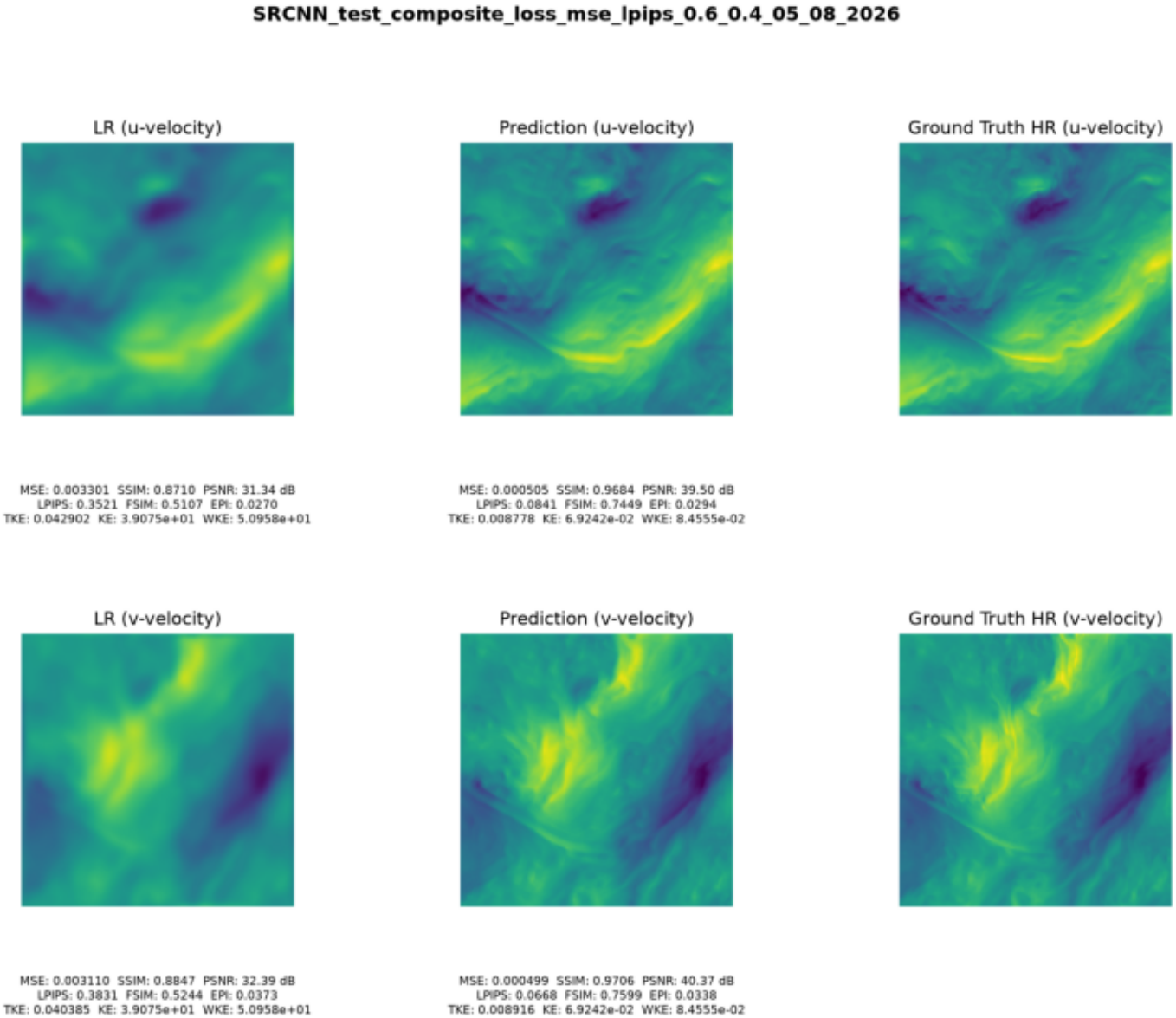




Top: kinetic-energy spectrum. Bottom: LR / Prediction / HR comparison. Both use the same fixed test-sample.

# SRCNN_test_composite_loss_mse_lpips_0.6_0.4_05_08_2026

0.6 MSE + 0.4 LPIPS

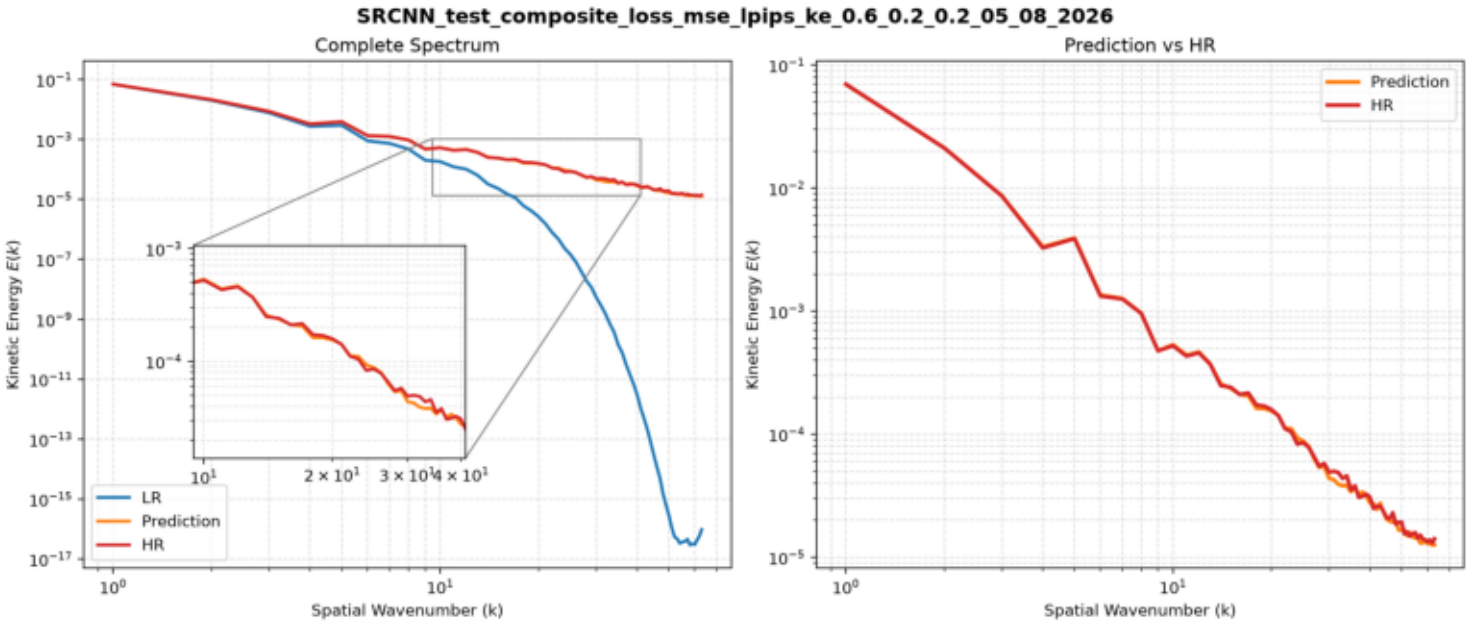


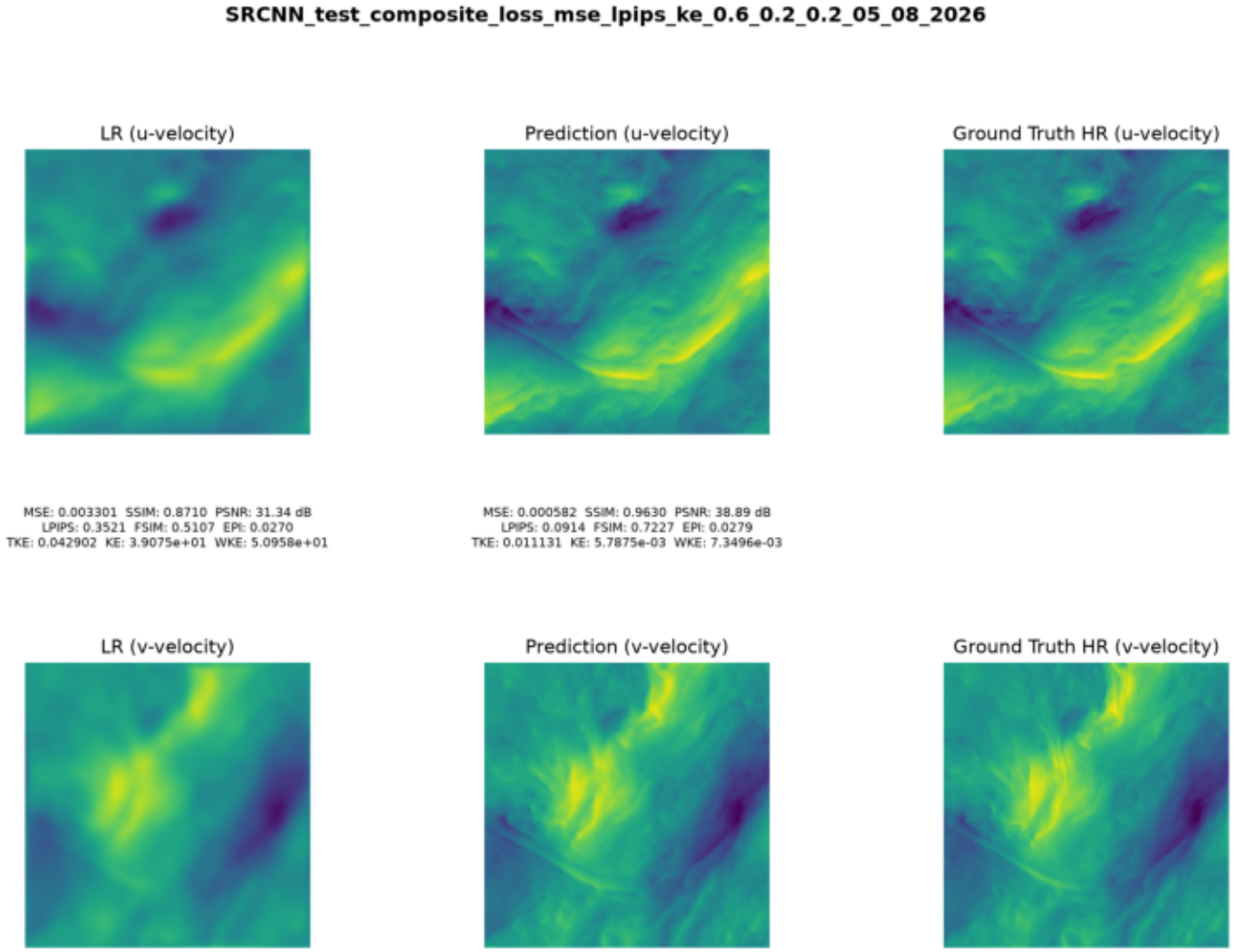




Top: kinetic-energy spectrum. Bottom: LR / Prediction / HR comparison. Both use the same fixed test-sample.

# SRCNN_test_composite_loss_mse_lpips_ke_0.6_0.2_0.2_05_08_2026

0.6 MSE + 0.2 LPIPS + 0.2 KE

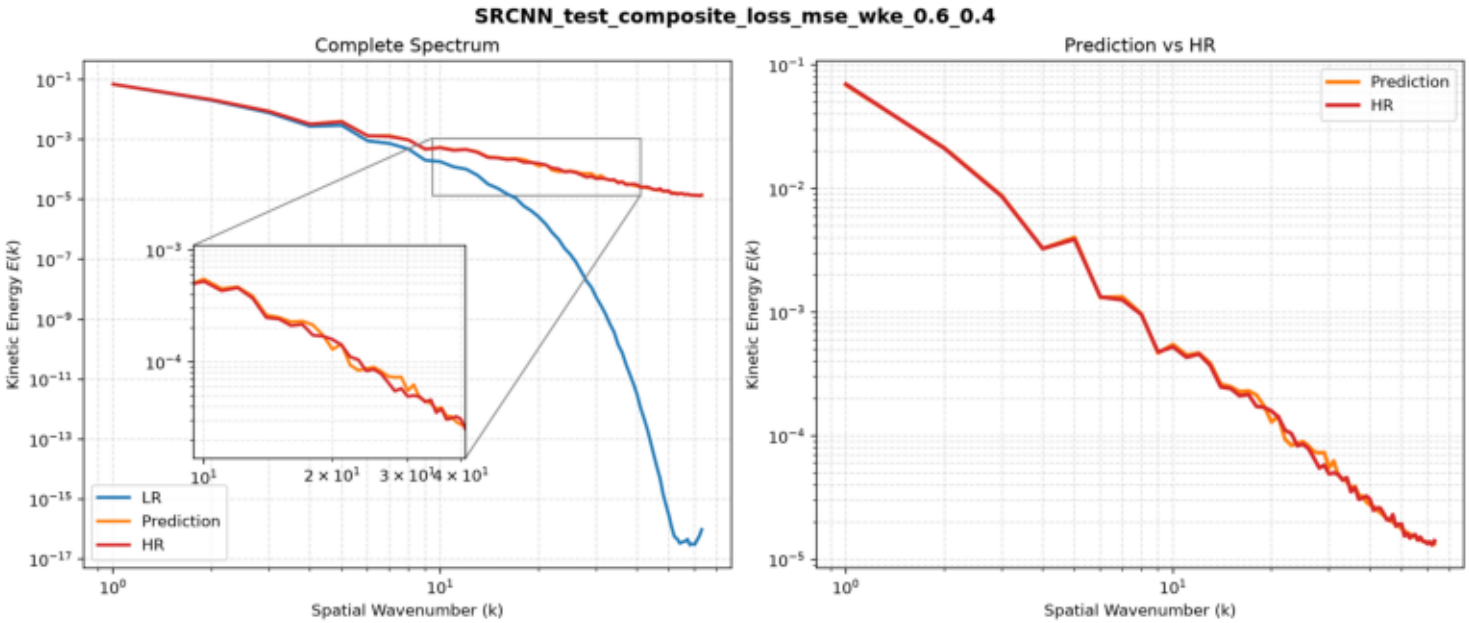


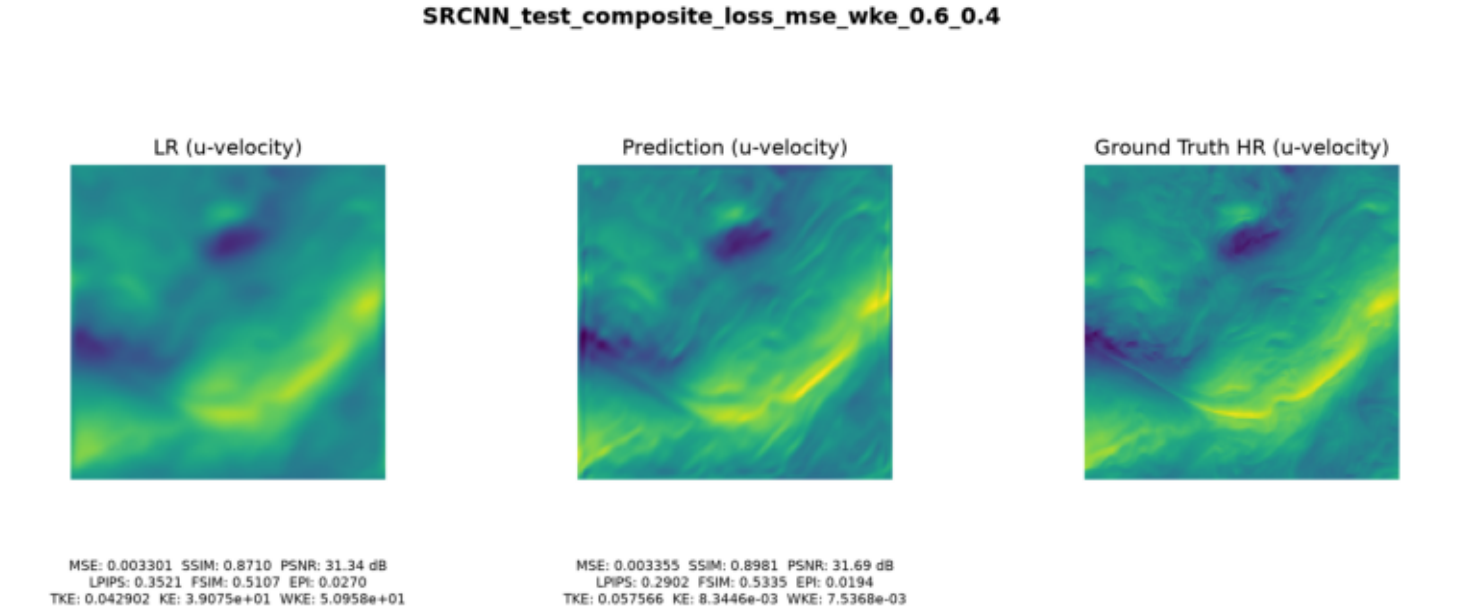




Top: kinetic-energy spectrum. Bottom: LR / Prediction / HR comparison. Both use the same fixed test-sample.

# SRCNN_test_composite_loss_mse_wke_0.6_0.4

0.6 MSE + 0.4 Weighted KE




Top: kinetic-energy spectrum. Bottom: LR / Prediction / HR comparison. Both use the same fixed test-sample.

# DSCMS -- Summary metrics (bold = best per column)

## u-velocity metrics

| Strategy | MSE | SSIM | PSNR | LPIPS | FSIM | EPI | TKE |
|---|---|---|---|---|---|---|---|
| MSE only | 0.0009604 | 0.9573 | 38.55 | 0.2036 | 0.6454 | **0.03413** | 0.01938 |
| 0.6 MSE + 0.4 LPIPS | **0.0002932** | **0.9873** | **43.73** | **0.03887** | **0.8244** | 0.02977 | **0.003927** |
| 0.6 MSE + 0.2 LPIPS + 0.2 KE | 0.0004624 | 0.9811 | 41.77 | 0.05766 | 0.7749 | 0.02898 | 0.00722 |
| 0.6 MSE + 0.4 Weighted KE | 0.002918 | 0.9181 | 33.83 | 0.2649 | 0.5555 | 0.02923 | 0.04498 |

## v-velocity metrics

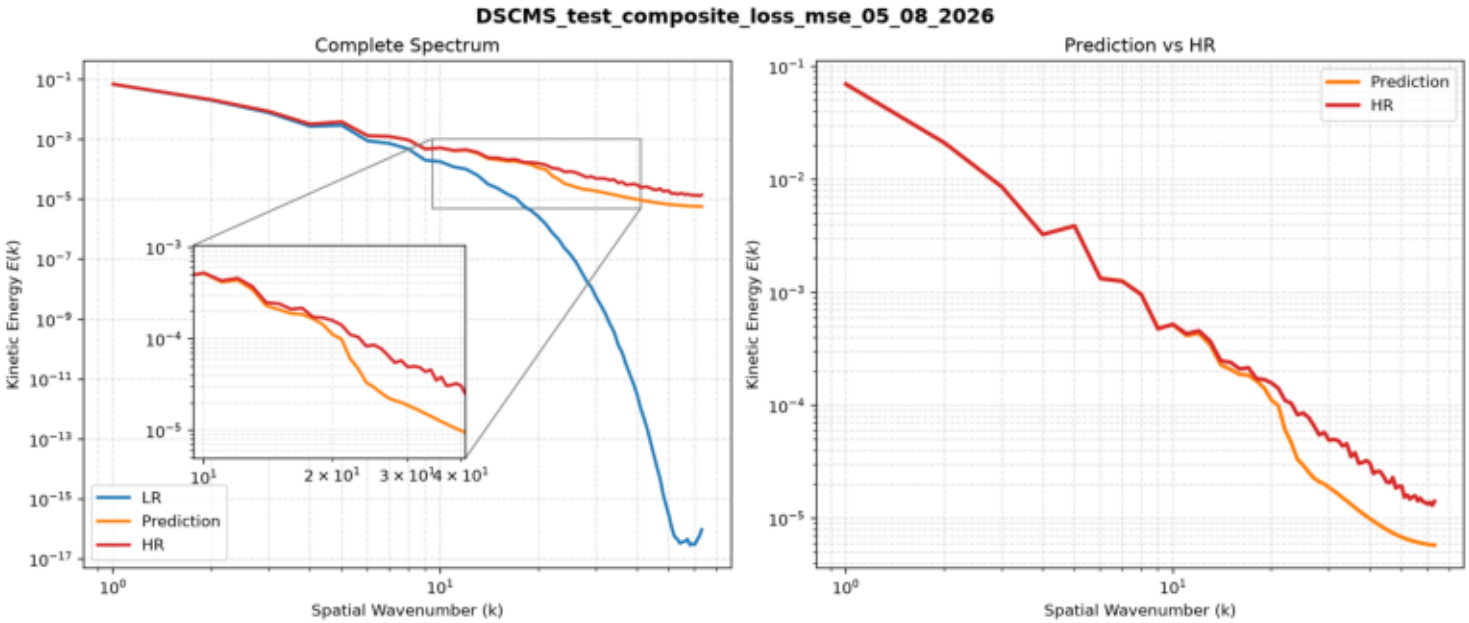
| Strategy | MSE | SSIM | PSNR | LPIPS | FSIM | EPI | TKE |
|---|---|---|---|---|---|---|---|
| MSE only | 0.001142 | 0.957 | 38.61 | 0.2084 | 0.7128 | **0.03014** | 0.02243 |
| 0.6 MSE + 0.4 LPIPS | **0.0003622** | **0.9885** | **43.6** | **0.03824** | **0.8639** | 0.02584 | **0.004476** |
| 0.6 MSE + 0.2 LPIPS + 0.2 KE | 0.0005142 | 0.9841 | 42.08 | 0.05325 | 0.8304 | 0.02553 | 0.007369 |
| 0.6 MSE + 0.4 Weighted KE | 0.003015 | 0.93 | 34.53 | 0.2679 | 0.6358 | 0.02634 | 0.04807 |

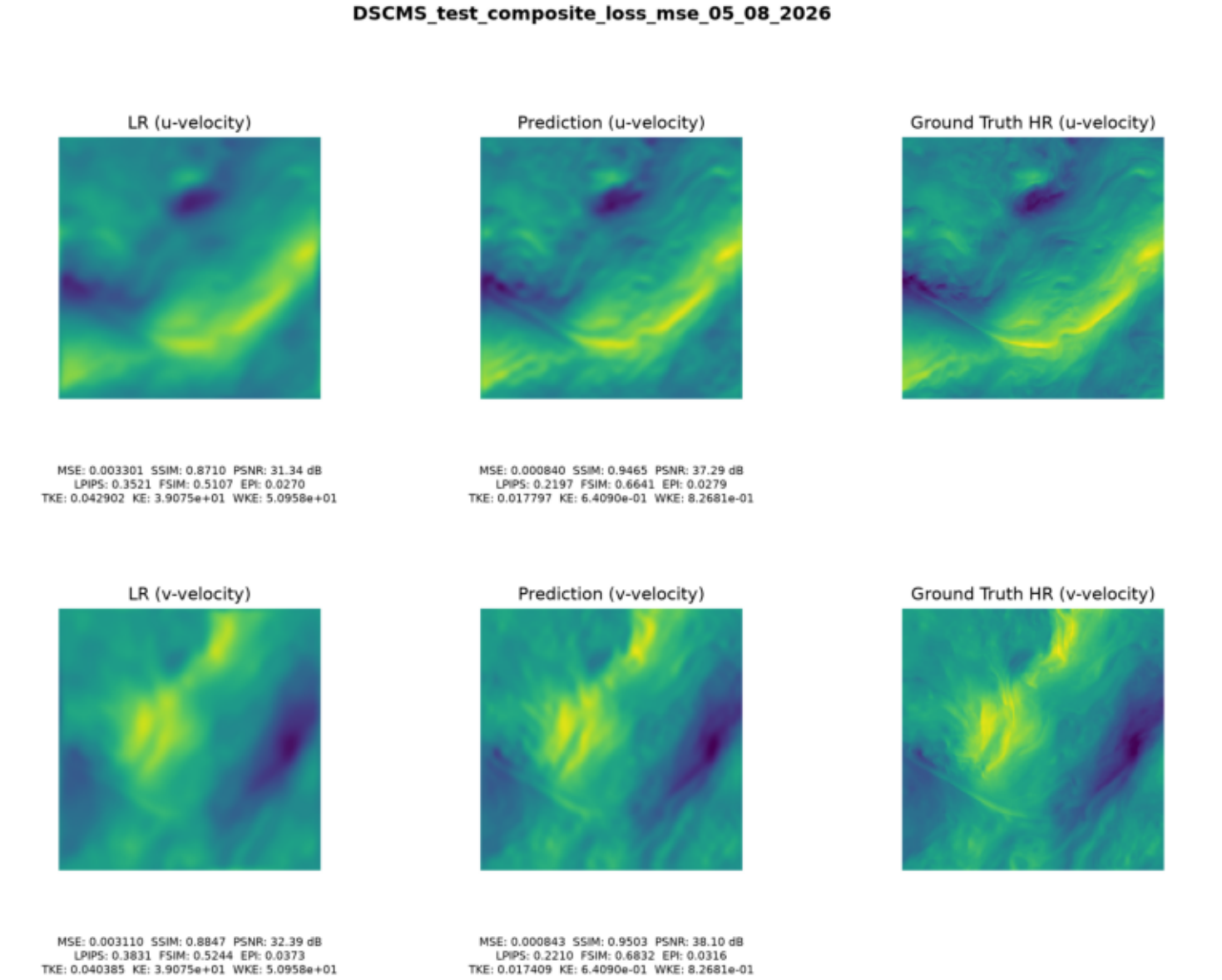
## Combined metrics (KE / WKE mix u and v, not split by channel)

| Strategy | KE | WKE |
|---|---|---|
| MSE only | 0.4309 | 0.5718 |
| 0.6 MSE + 0.4 LPIPS | 0.02249 | 0.03378 |
| 0.6 MSE + 0.2 LPIPS + 0.2 KE | **0.002928** | **0.004064** |
| 0.6 MSE + 0.4 Weighted KE | 0.01021 | 0.01143 |

# DSCMS_test_composite_loss_mse_05_08_2026

MSE only

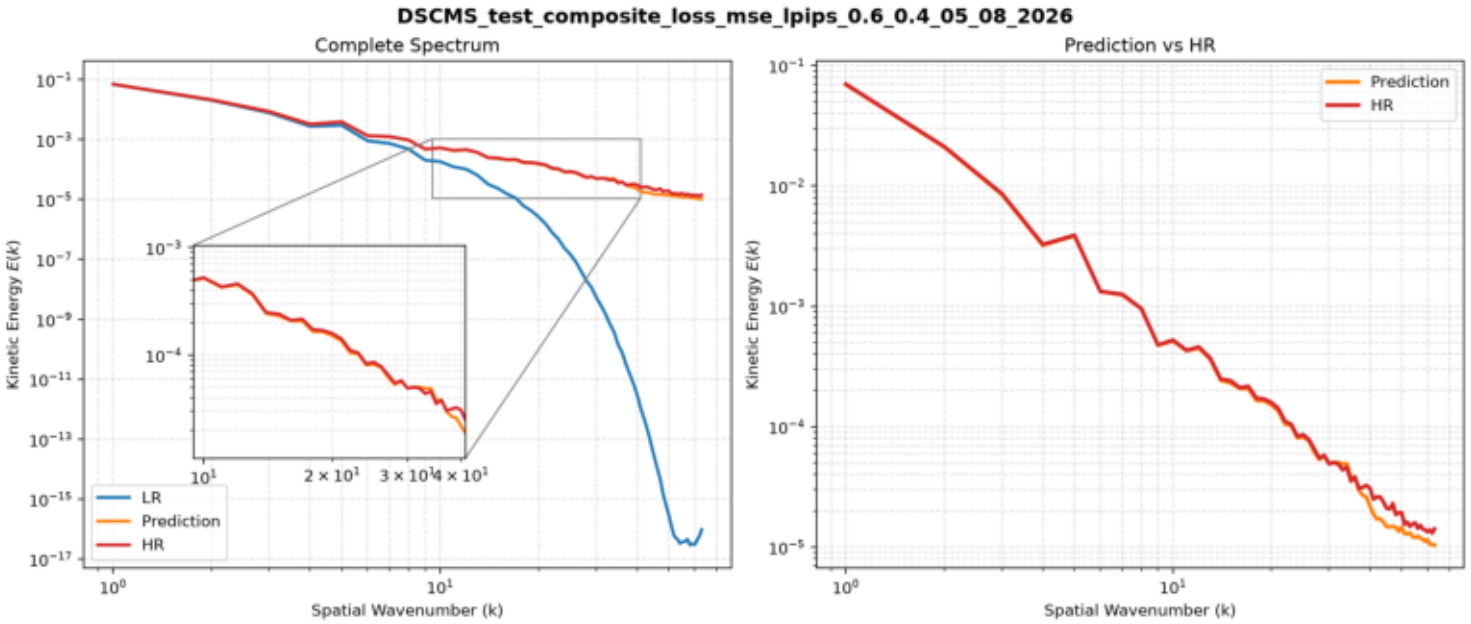


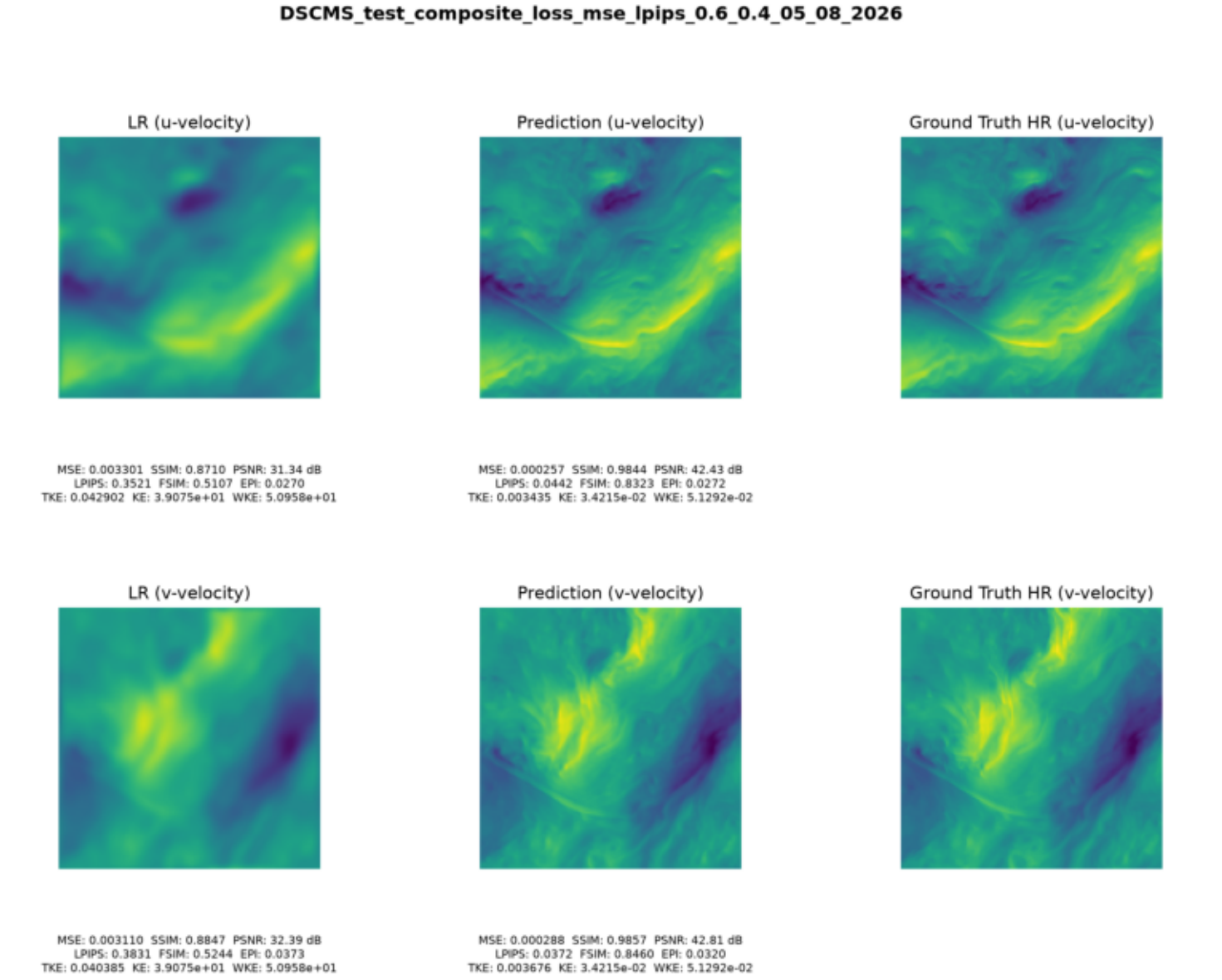




Top: kinetic-energy spectrum. Bottom: LR / Prediction / HR comparison. Both use the same fixed test-sample.

# DSCMS_test_composite_loss_mse_lpips_0.6_0.4_05_08_2026

0.6 MSE + 0.4 LPIPS

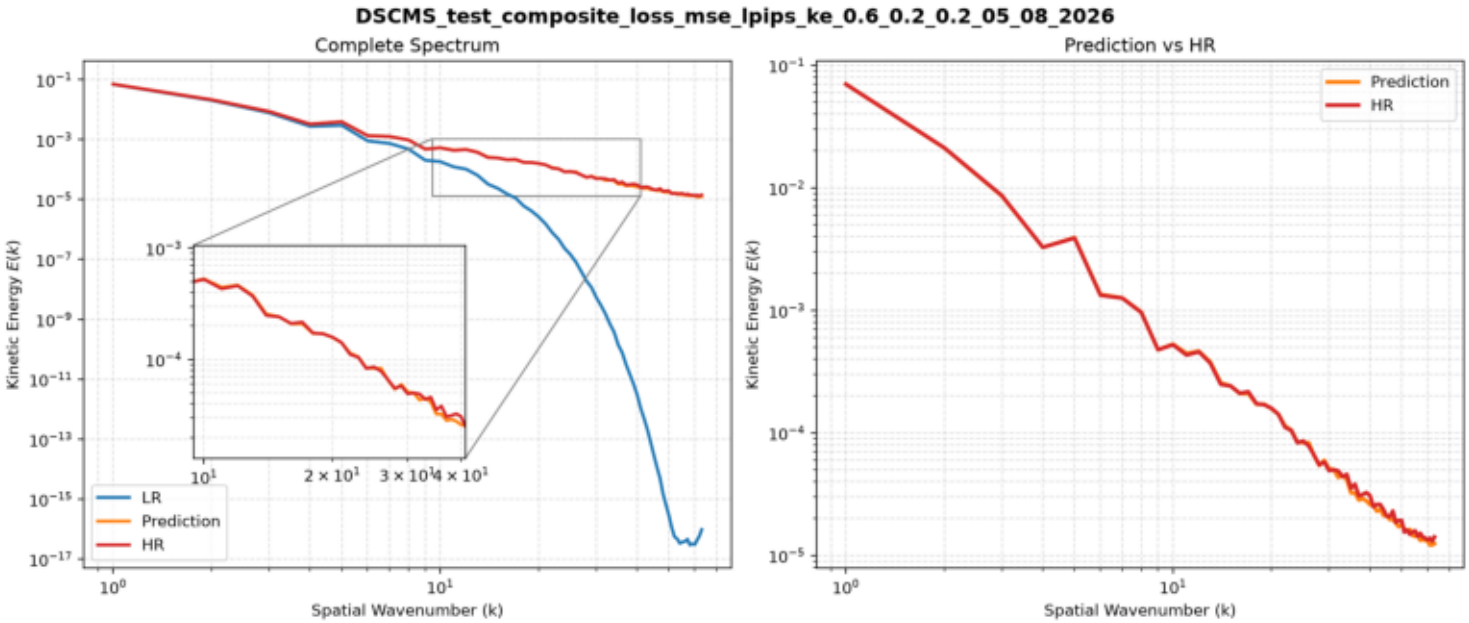


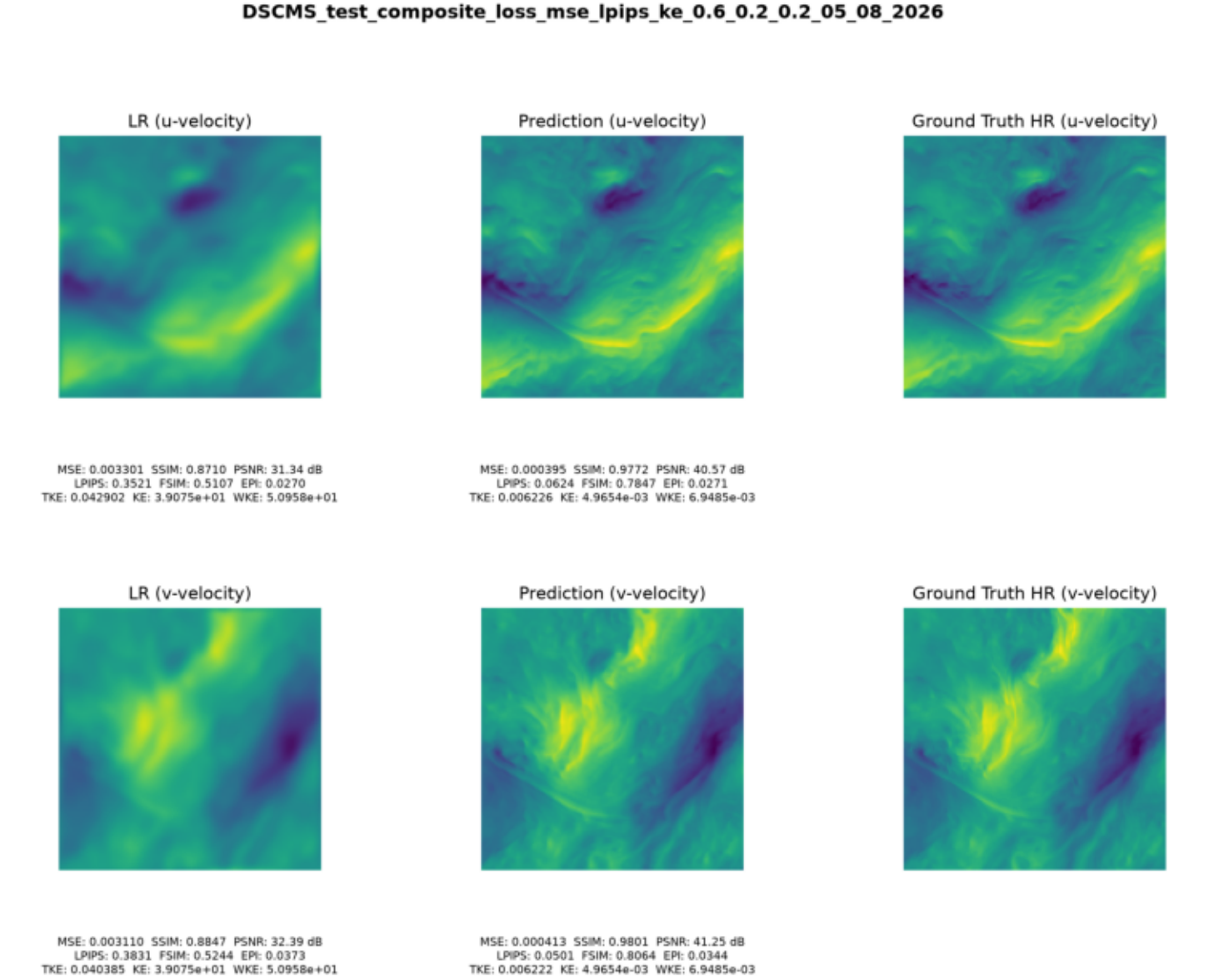




Top: kinetic-energy spectrum. Bottom: LR / Prediction / HR comparison. Both use the same fixed test-sample.

# DSCMS_test_composite_loss_mse_lpips_ke_0.6_0.2_0.2_05_08_2026

0.6 MSE + 0.2 LPIPS + 0.2 KE

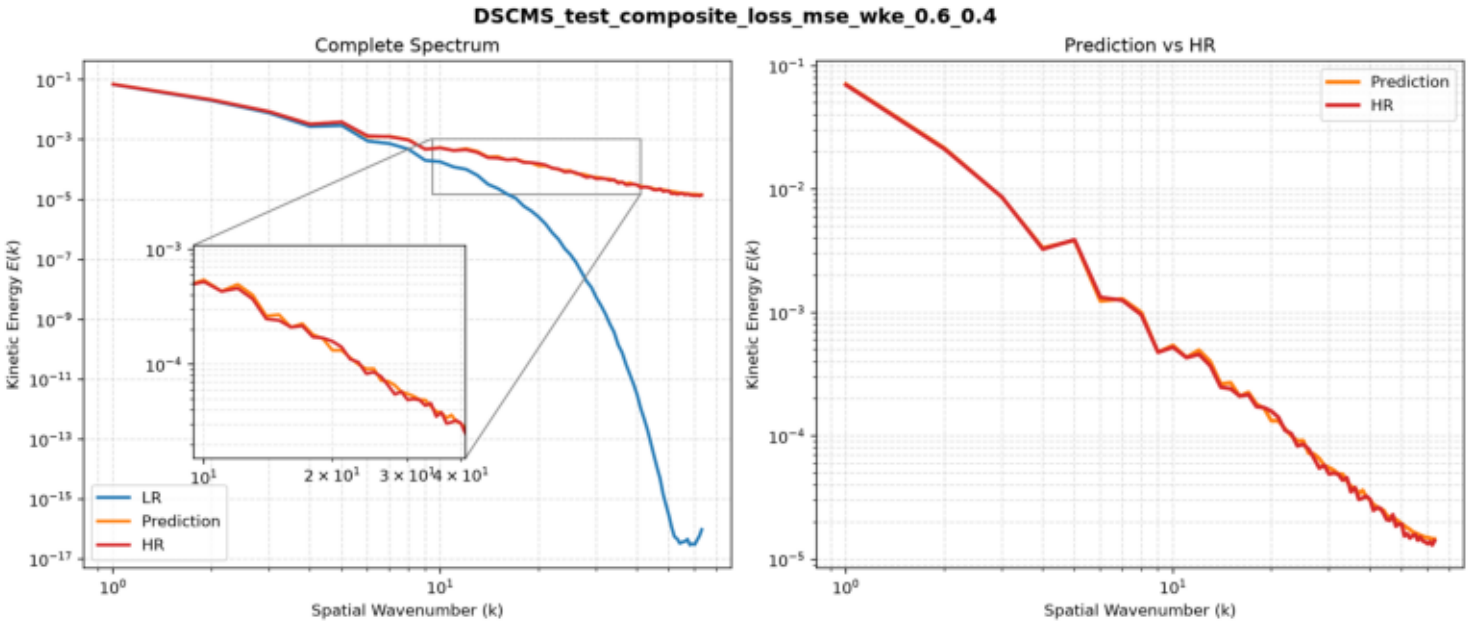


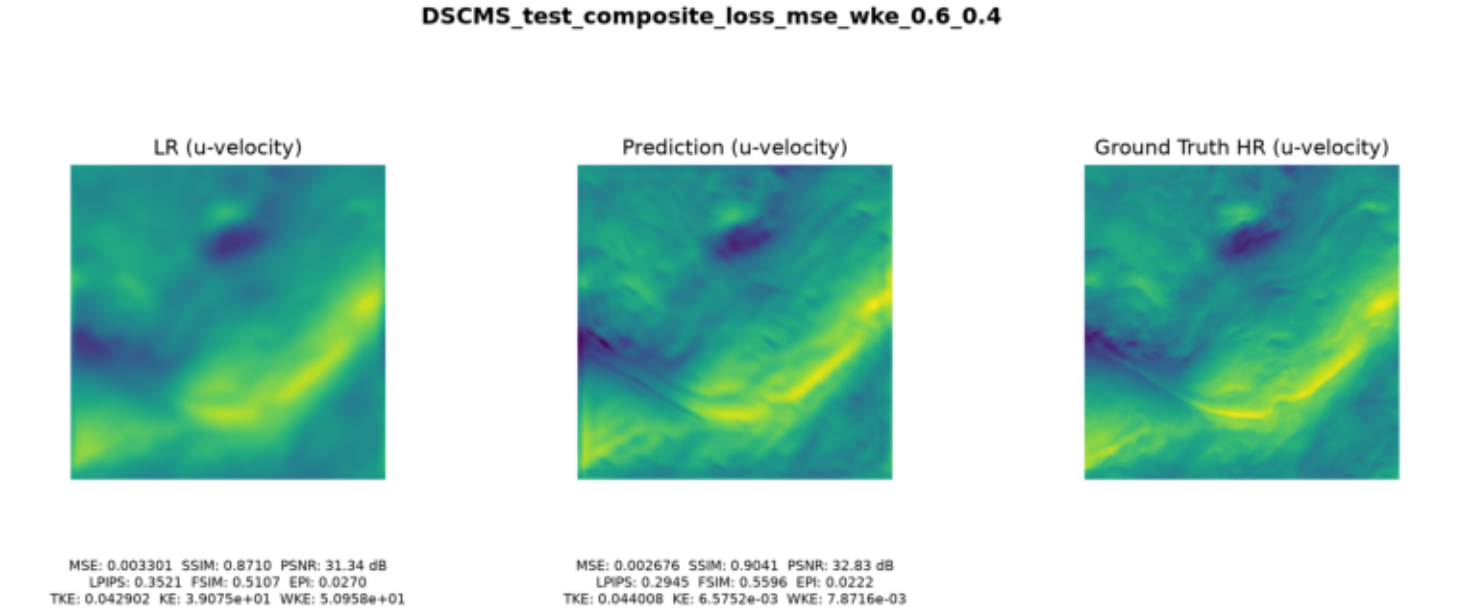




Top: kinetic-energy spectrum. Bottom: LR / Prediction / HR comparison. Both use the same fixed test-sample.

# DSCMS_test_composite_loss_mse_wke_0.6_0.4

0.6 MSE + 0.4 Weighted KE




Top: kinetic-energy spectrum. Bottom: LR / Prediction / HR comparison. Both use the same fixed test-sample.

# PRUSR -- Summary metrics (bold = best per column)

## u-velocity metrics

| Strategy | MSE | SSIM | PSNR | LPIPS | FSIM | EPI | TKE |
|---|---|---|---|---|---|---|---|
| MSE only | 0.0007929 | 0.9618 | 39.41 | 0.1877 | 0.6633 | **0.03304** | 0.01667 |
| 0.6 MSE + 0.4 LPIPS | **0.0003751** | **0.9807** | **42.61** | **0.04372** | **0.7976** | 0.02965 | **0.004924** |
| 0.6 MSE + 0.2 LPIPS + 0.2 KE | 0.0004483 | 0.9788 | 41.89 | 0.0552 | 0.7651 | 0.02893 | 0.007009 |
| 0.6 MSE + 0.4 Weighted KE | 0.001657 | 0.9492 | 36.28 | 0.211 | 0.6194 | 0.02802 | 0.02837 |

## v-velocity metrics

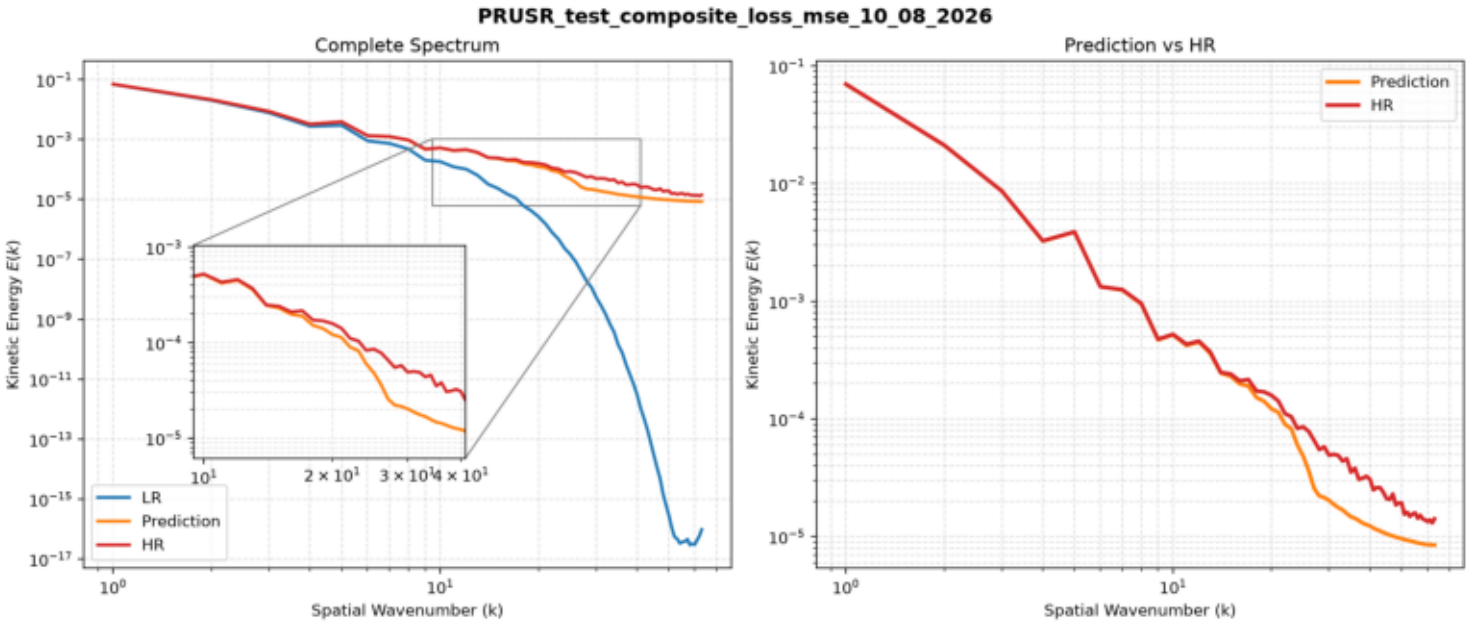
| Strategy | MSE | SSIM | PSNR | LPIPS | FSIM | EPI | TKE |
|---|---|---|---|---|---|---|---|
| MSE only | 0.0008583 | 0.9648 | 39.89 | 0.1838 | 0.7377 | **0.02883** | 0.01782 |
| 0.6 MSE + 0.4 LPIPS | **0.0003447** | **0.9852** | **43.82** | **0.03762** | **0.8603** | 0.0263 | **0.004319** |
| 0.6 MSE + 0.2 LPIPS + 0.2 KE | 0.000466 | 0.9811 | 42.55 | 0.05191 | 0.8217 | 0.02555 | 0.007244 |
| 0.6 MSE + 0.4 Weighted KE | 0.001837 | 0.956 | 36.67 | 0.2055 | 0.7048 | 0.02502 | 0.02976 |

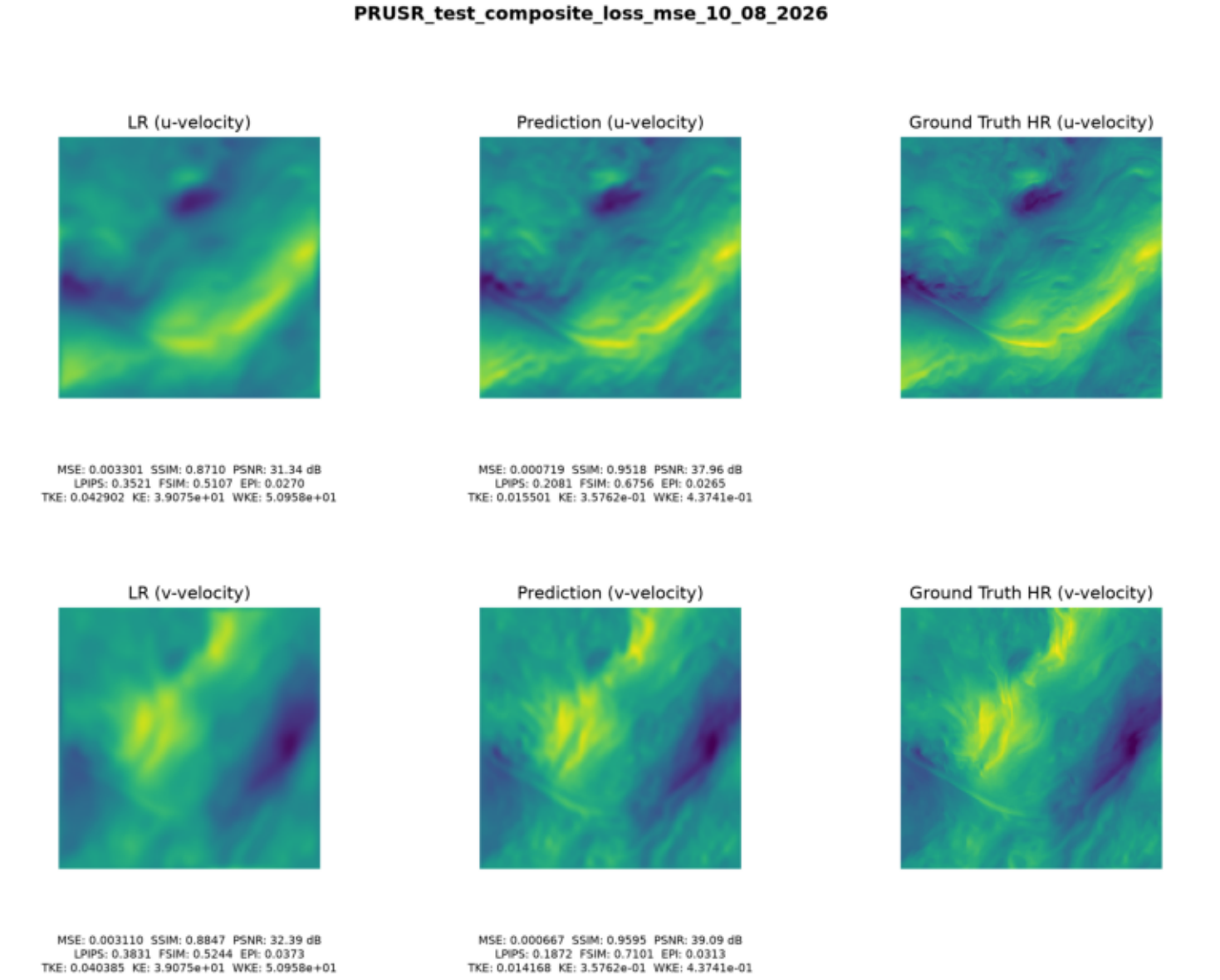
## Combined metrics (KE / WKE mix u and v, not split by channel)

| Strategy | KE | WKE |
|---|---|---|
| MSE only | 0.2072 | 0.2534 |
| 0.6 MSE + 0.4 LPIPS | 0.02035 | 0.02915 |
| 0.6 MSE + 0.2 LPIPS + 0.2 KE | **0.002722** | **0.003771** |
| 0.6 MSE + 0.4 Weighted KE | 0.007643 | 0.009628 |

# PRUSR_test_composite_loss_mse_10_08_2026

MSE only

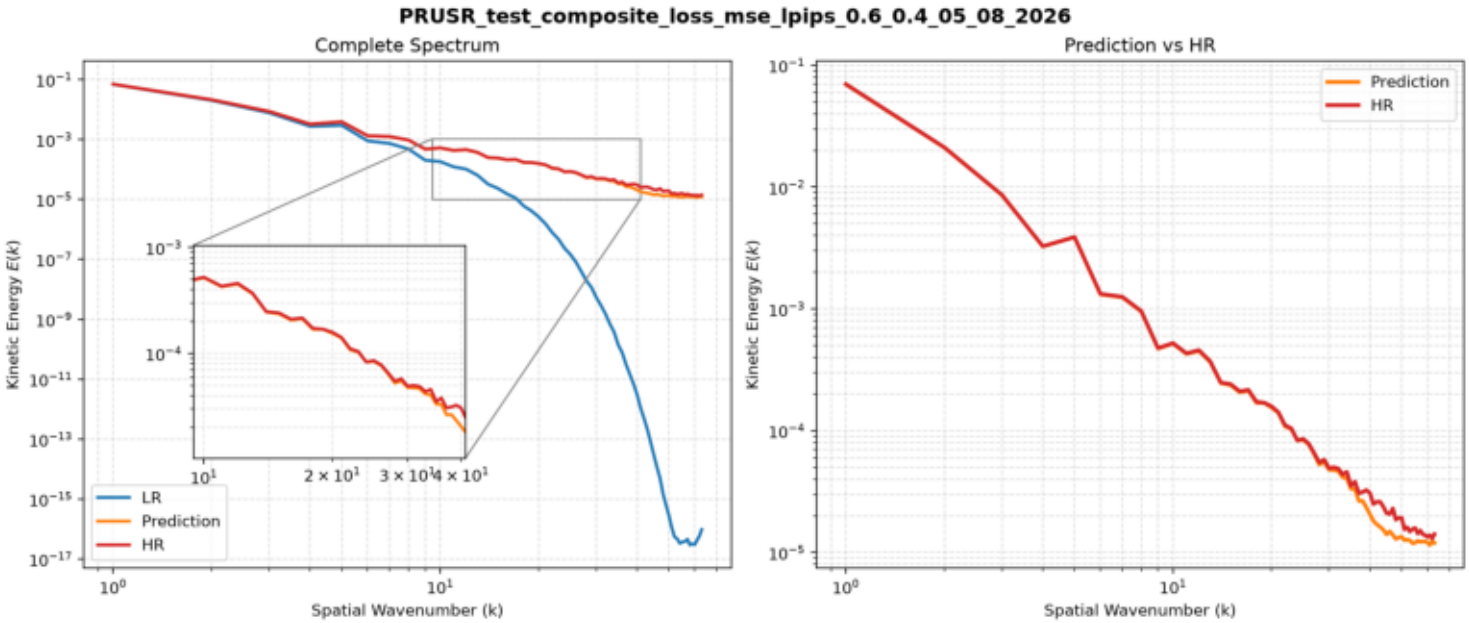


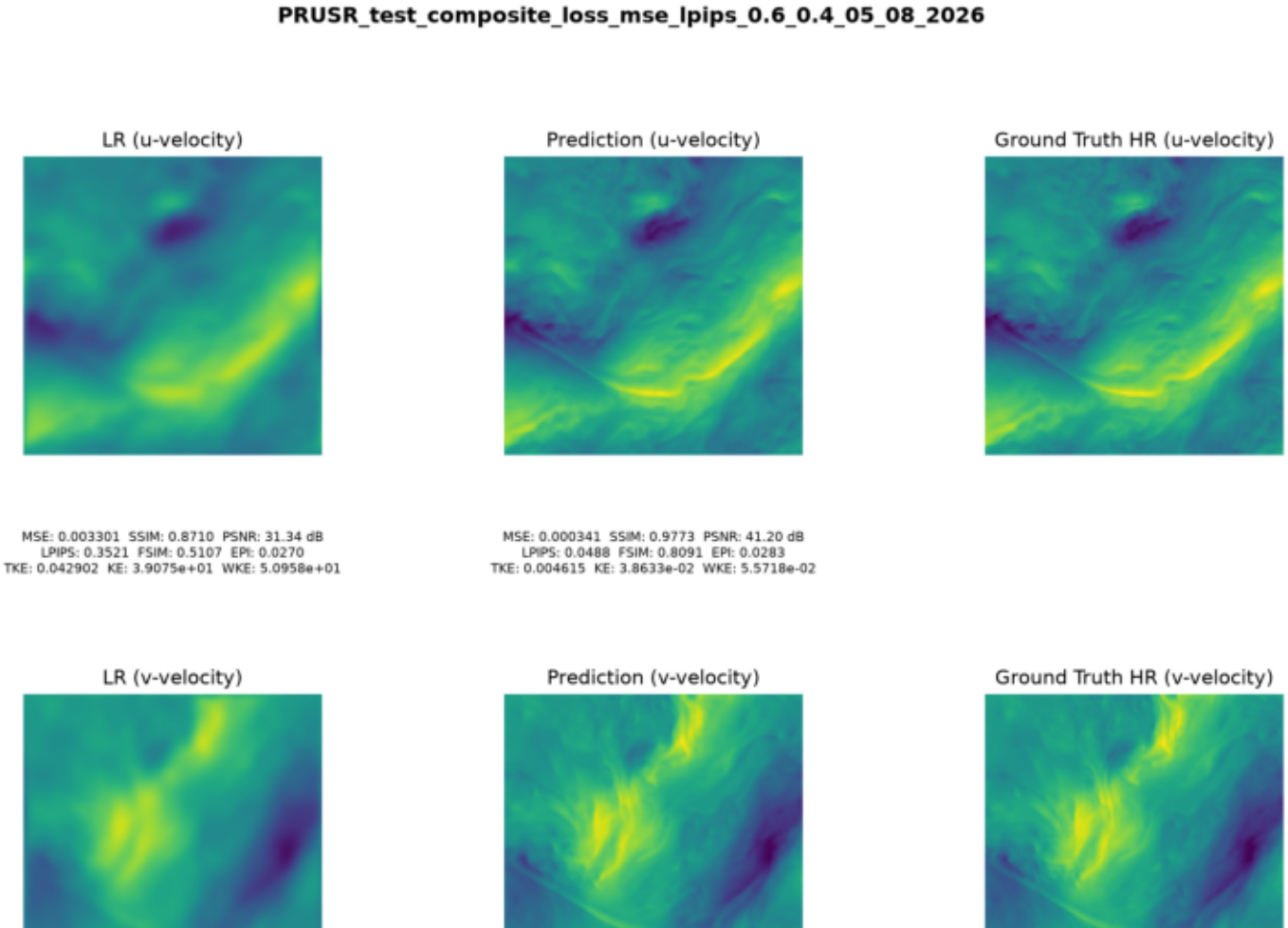




Top: kinetic-energy spectrum. Bottom: LR / Prediction / HR comparison. Both use the same fixed test-sample.

# PRUSR_test_composite_loss_mse_lpips_0.6_0.4_05_08_2026

0.6 MSE + 0.4 LPIPS

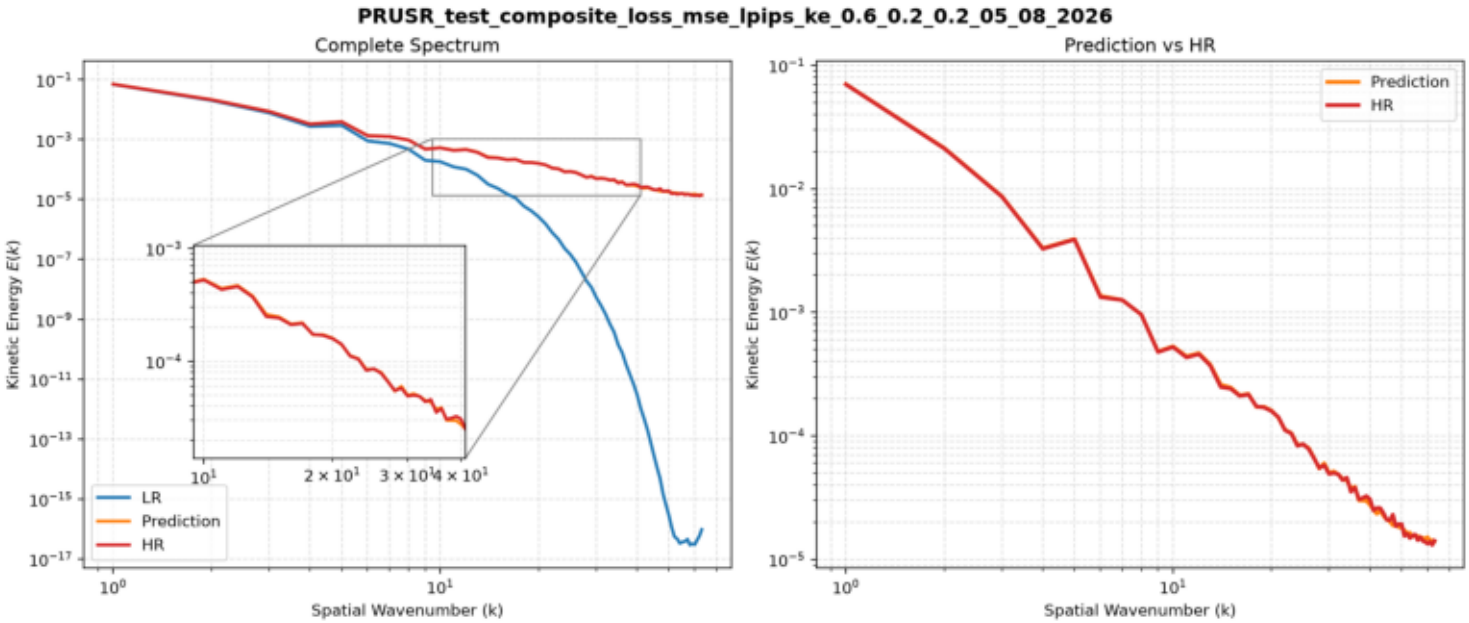


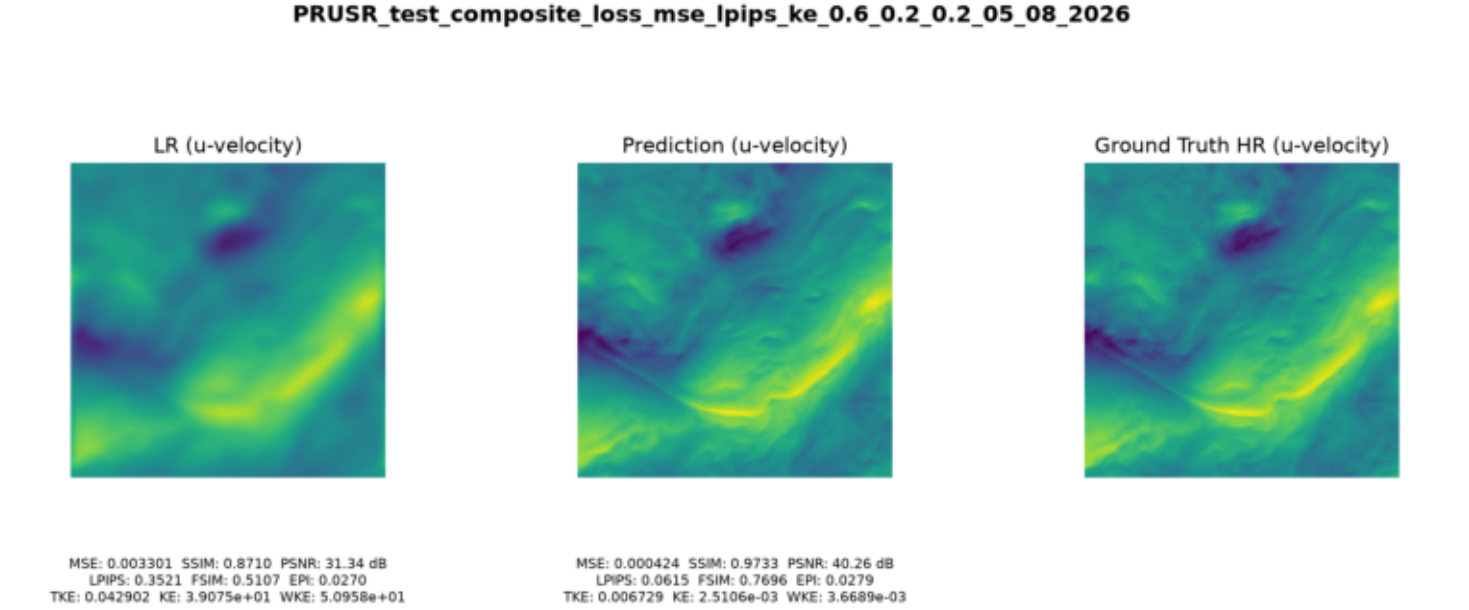




Top: kinetic-energy spectrum. Bottom: LR / Prediction / HR comparison. Both use the same fixed test-sample.

# PRUSR_test_composite_loss_mse_lpips_ke_0.6_0.2_0.2_05_08_2026

0.6 MSE + 0.2 LPIPS + 0.2 KE

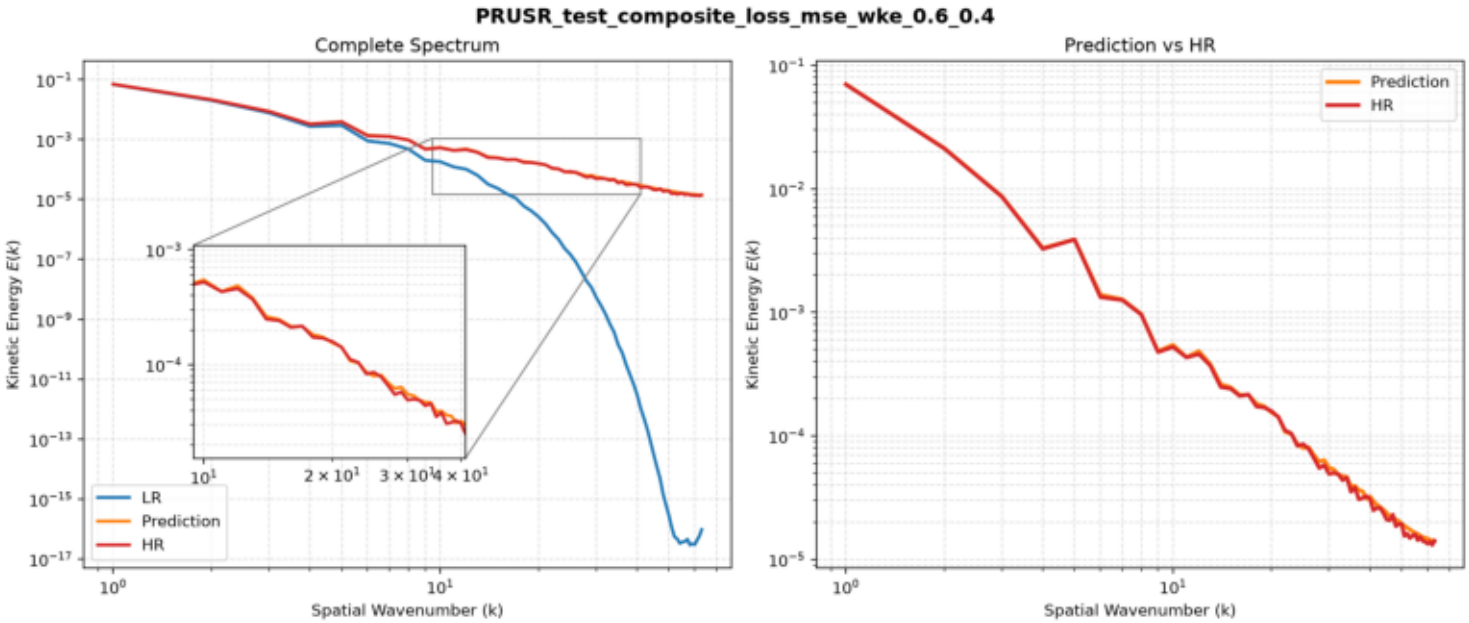


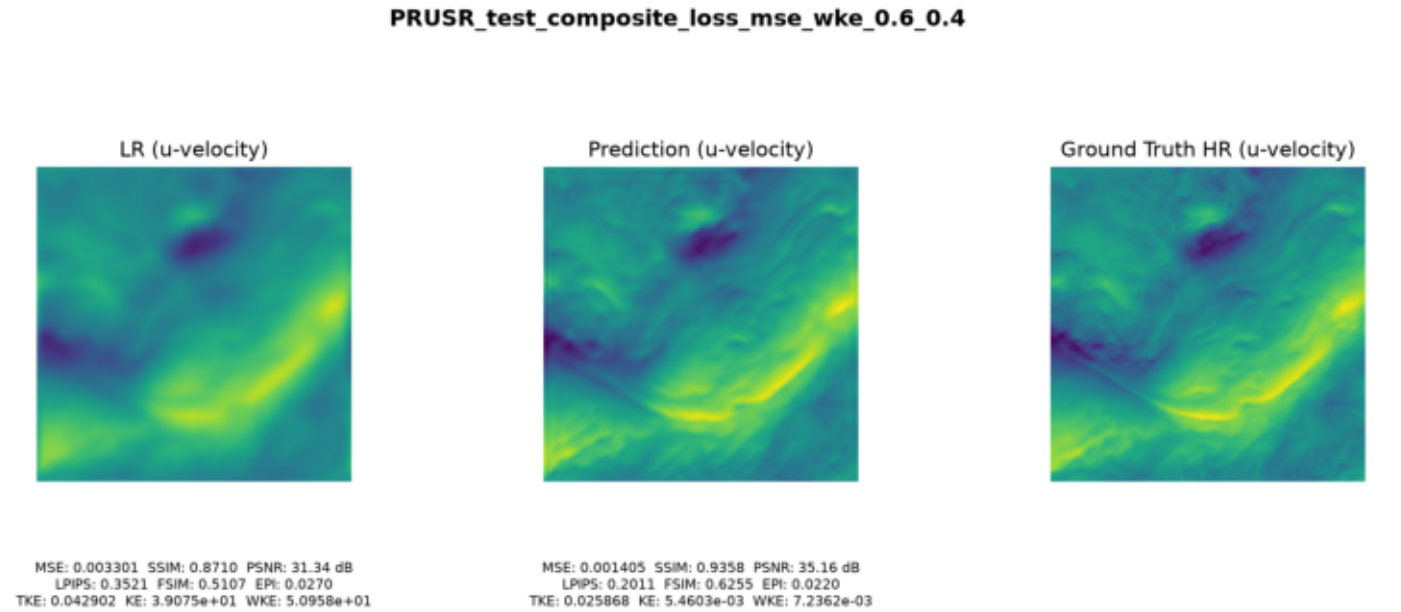




Top: kinetic-energy spectrum. Bottom: LR / Prediction / HR comparison. Both use the same fixed test-sample.

# PRUSR_test_composite_loss_mse_wke_0.6_0.4

0.6 MSE + 0.4 Weighted KE




Top: kinetic-energy spectrum. Bottom: LR / Prediction / HR comparison. Both use the same fixed test-sample.

# SoFoNO -- Summary metrics (bold = best per column)

## u-velocity metrics

| Strategy | MSE | SSIM | PSNR | LPIPS | FSIM | EPI | TKE |
|---|---|---|---|---|---|---|---|
| MSE only | 0.0006556 | 0.9702 | 40.23 | 0.1673 | 0.6975 | **0.03203** | 0.01284 |
| 0.6 MSE + 0.4 LPIPS | **0.000115** | **0.9952** | **47.79** | **0.01328** | **0.9131** | 0.03008 | **0.0008854** |
| 0.6 MSE + 0.2 LPIPS + 0.2 KE | 0.0001373 | 0.9946 | 47.05 | 0.01583 | 0.9025 | 0.03014 | 0.00117 |
| 0.6 MSE + 0.4 Weighted KE | 0.0008592 | 0.9713 | 39.21 | 0.1323 | 0.6914 | 0.02959 | 0.01537 |

## v-velocity metrics

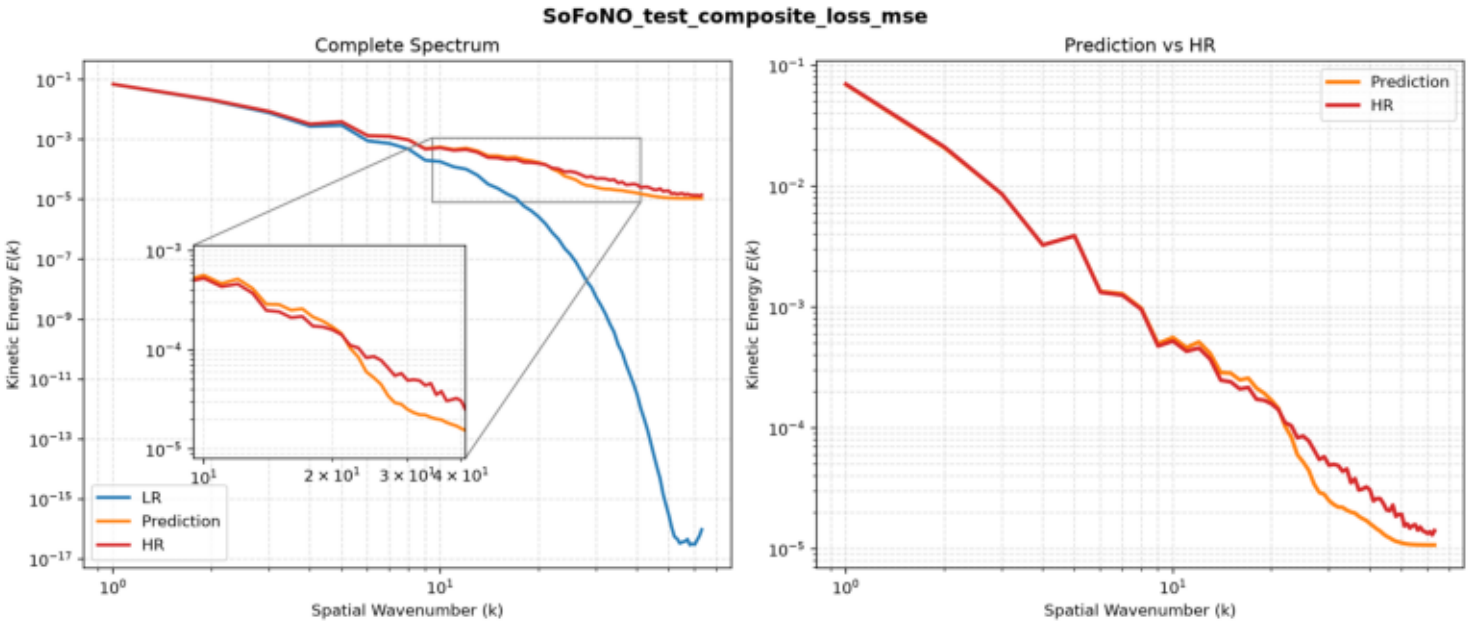
| Strategy | MSE | SSIM | PSNR | LPIPS | FSIM | EPI | TKE |
|---|---|---|---|---|---|---|---|
| MSE only | 0.000672 | 0.9744 | 40.95 | 0.1539 | 0.7738 | **0.02779** | 0.01306 |
| 0.6 MSE + 0.4 LPIPS | **0.0001225** | **0.9956** | **48.34** | **0.01291** | **0.9356** | 0.02651 | **0.0009511** |
| 0.6 MSE + 0.2 LPIPS + 0.2 KE | 0.000156 | 0.9951 | 47.32 | 0.01522 | 0.928 | 0.02662 | 0.00127 |
| 0.6 MSE + 0.4 Weighted KE | 0.0007128 | 0.9759 | 40.7 | 0.124 | 0.7788 | 0.0263 | 0.01332 |

## Combined metrics (KE / WKE mix u and v, not split by channel)

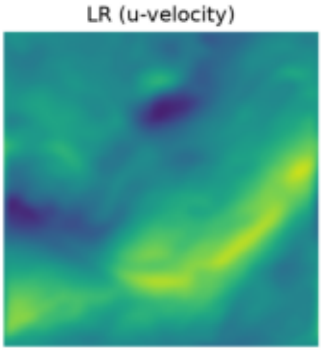
| Strategy | KE | WKE |
|---|---|---|
| MSE only | 0.1325 | 0.156 |
| 0.6 MSE + 0.4 LPIPS | 0.006835 | 0.01126 |
| 0.6 MSE + 0.2 LPIPS + 0.2 KE | **0.001097** | **0.001691** |
| 0.6 MSE + 0.4 Weighted KE | 0.002858 | 0.003615 |

# SoFoNO_test_composite_loss_mse

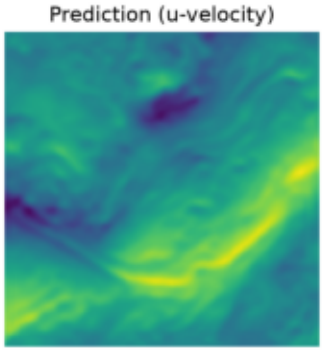
MSE only



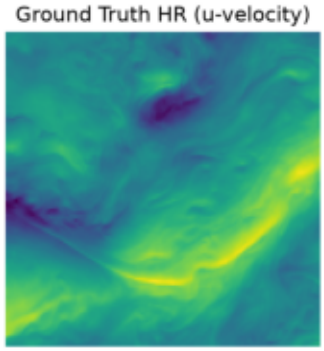

**SoFoNO_test_composite_loss_mse**

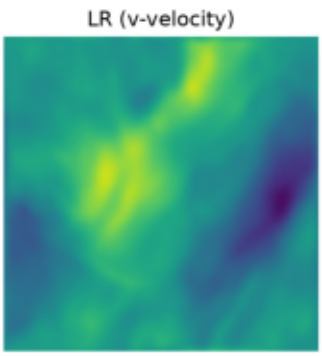



MSE: 0.003301 SSIM: 0.8710 PSNR: 31.34 dB
LPIPS: 0.3521 FSIM: 0.5107 EPI: 0.0270
TKE: 0.042902 KE: 3.9075e+01 WKE: 5.0958e+01

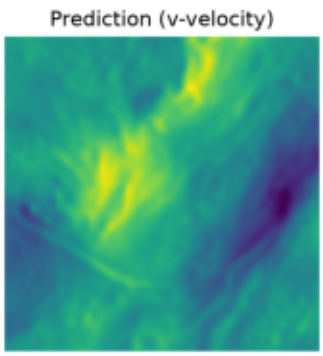



MSE: 0.000563 SSIM: 0.9635 PSNR: 39.02 dB
LPIPS: 0.1890 FSIM: 0.7161 EPI: 0.0282
TKE: 0.011322 KE: 1.8041e-01 WKE: 2.1399e-01

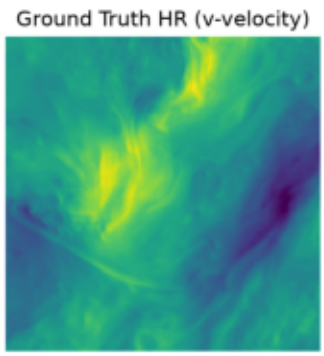



MSE: 0.003110 SSIM: 0.8847 PSNR: 32.39 dB
LPIPS: 0.3831 FSIM: 0.5244 EPI: 0.0373
TKE: 0.040385 KE: 3.9075e+01 WKE: 5.0958e+01



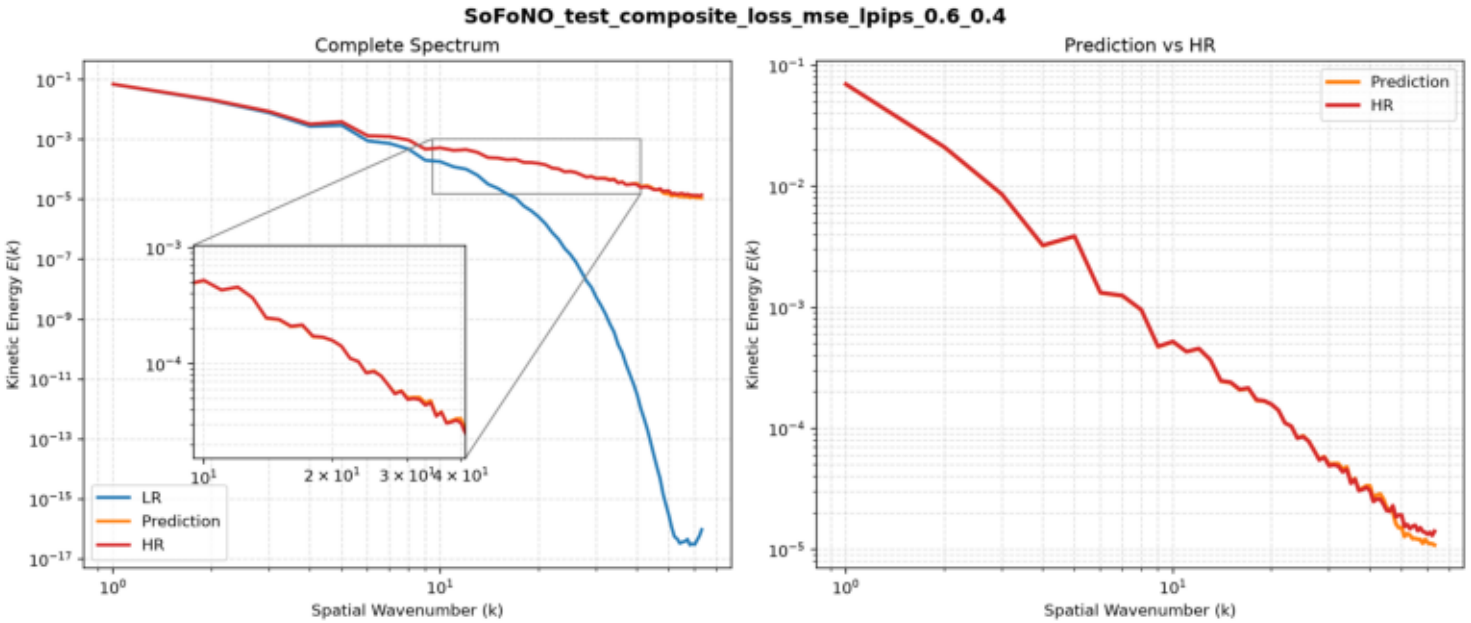

MSE: 0.000537 SSIM: 0.9695 PSNR: 40.03 dB
LPIPS: 0.1587 FSIM: 0.7455 EPI: 0.0326
TKE: 0.010644 KE: 1.8041e-01 WKE: 2.1399e-01

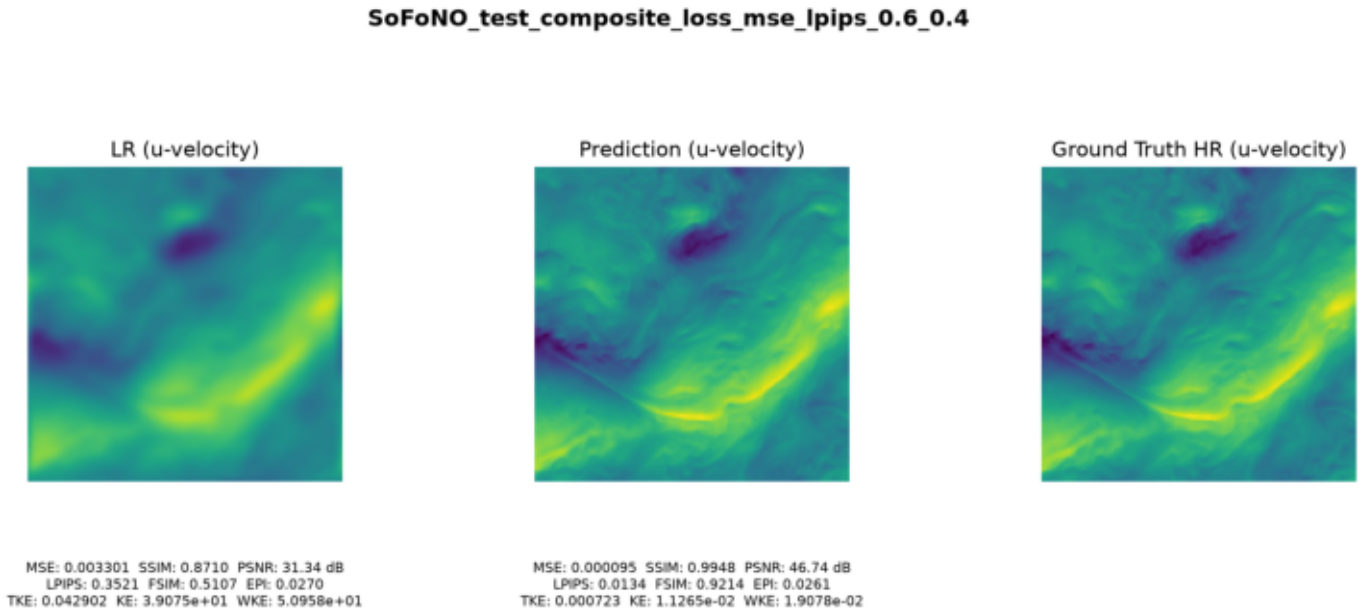



Top: kinetic-energy spectrum. Bottom: LR / Prediction / HR comparison. Both use the same fixed test-sample.

# SoFoNO_test_composite_loss_mse_lpips_0.6_0.4

0.6 MSE + 0.4 LPIPS

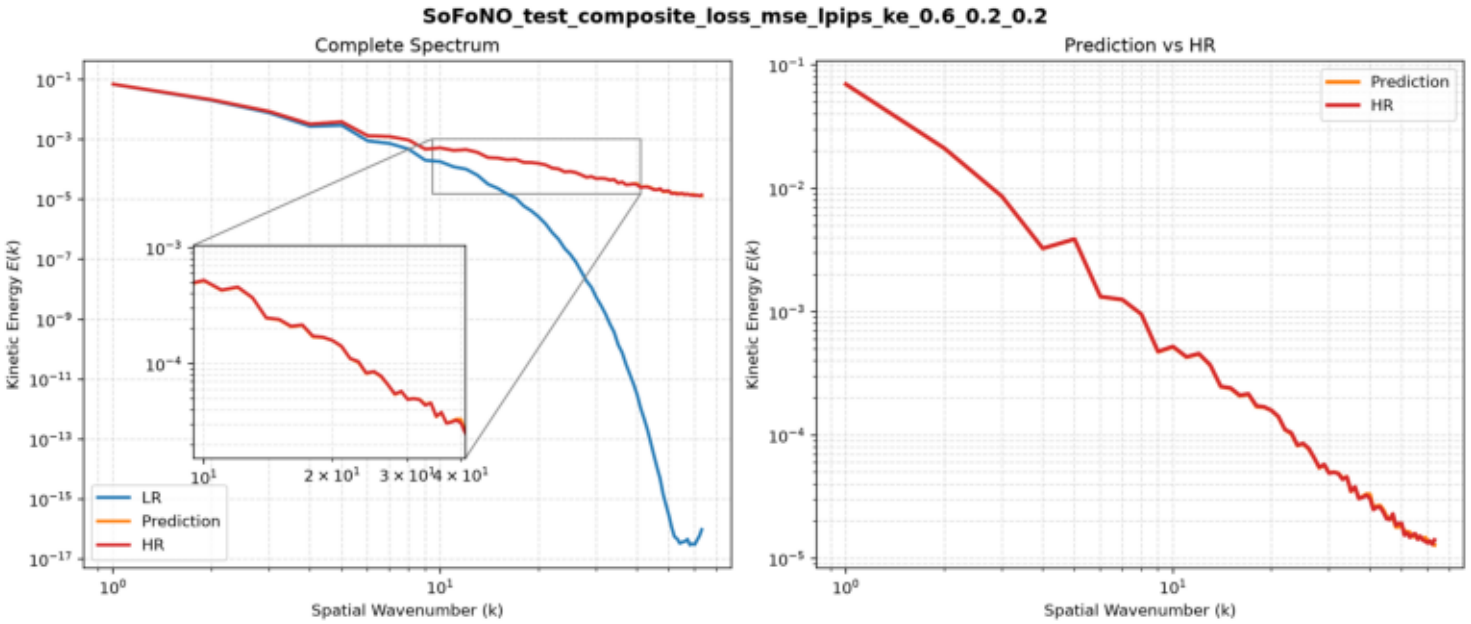


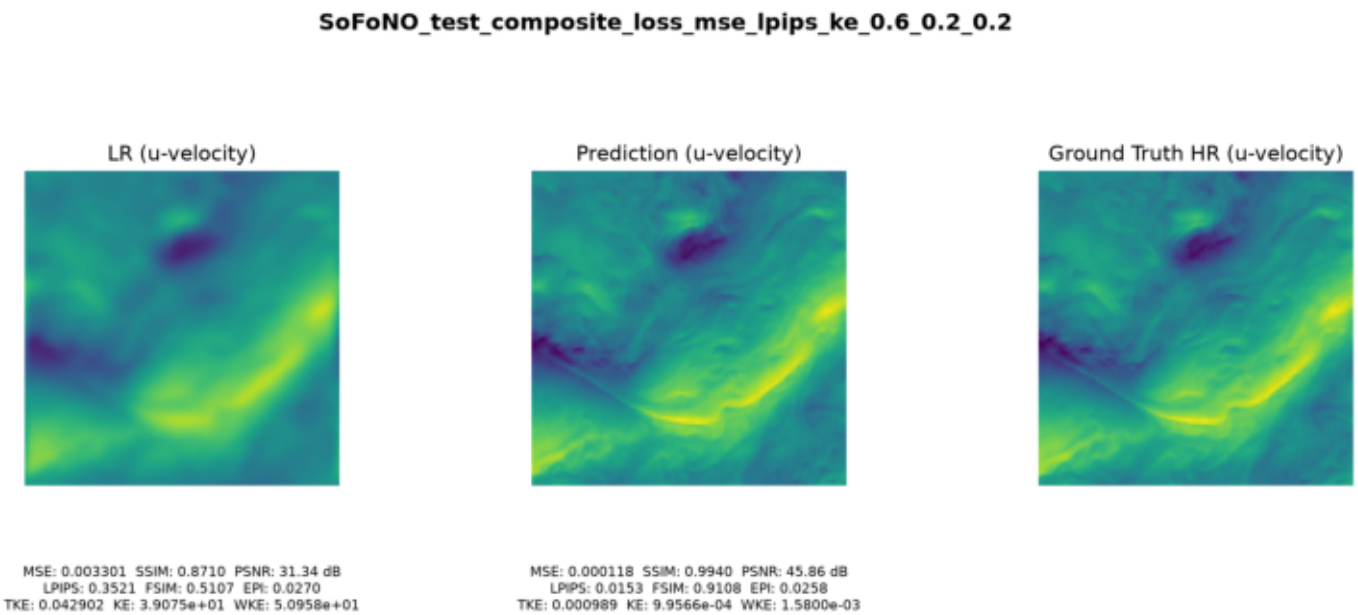




Top: kinetic-energy spectrum. Bottom: LR / Prediction / HR comparison. Both use the same fixed test-sample.

# SoFoNO_test_composite_loss_mse_lpips_ke_0.6_0.2_0.2

0.6 MSE + 0.2 LPIPS + 0.2 KE

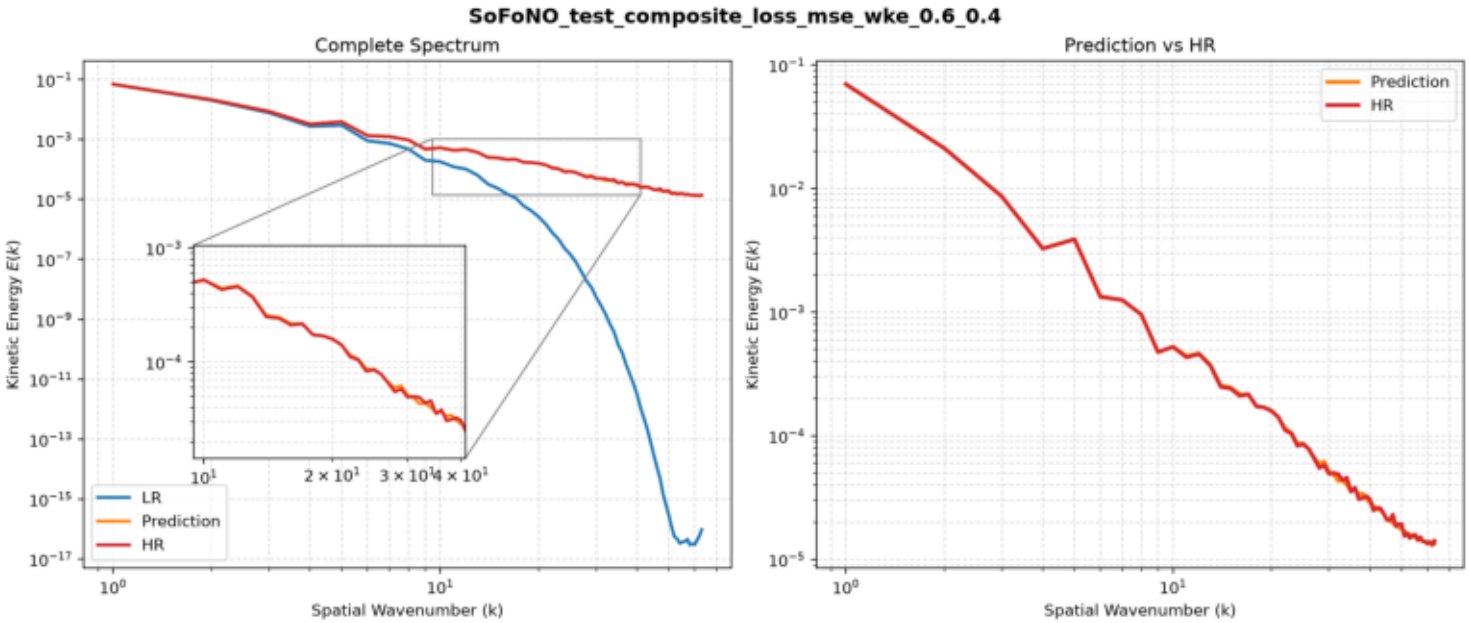


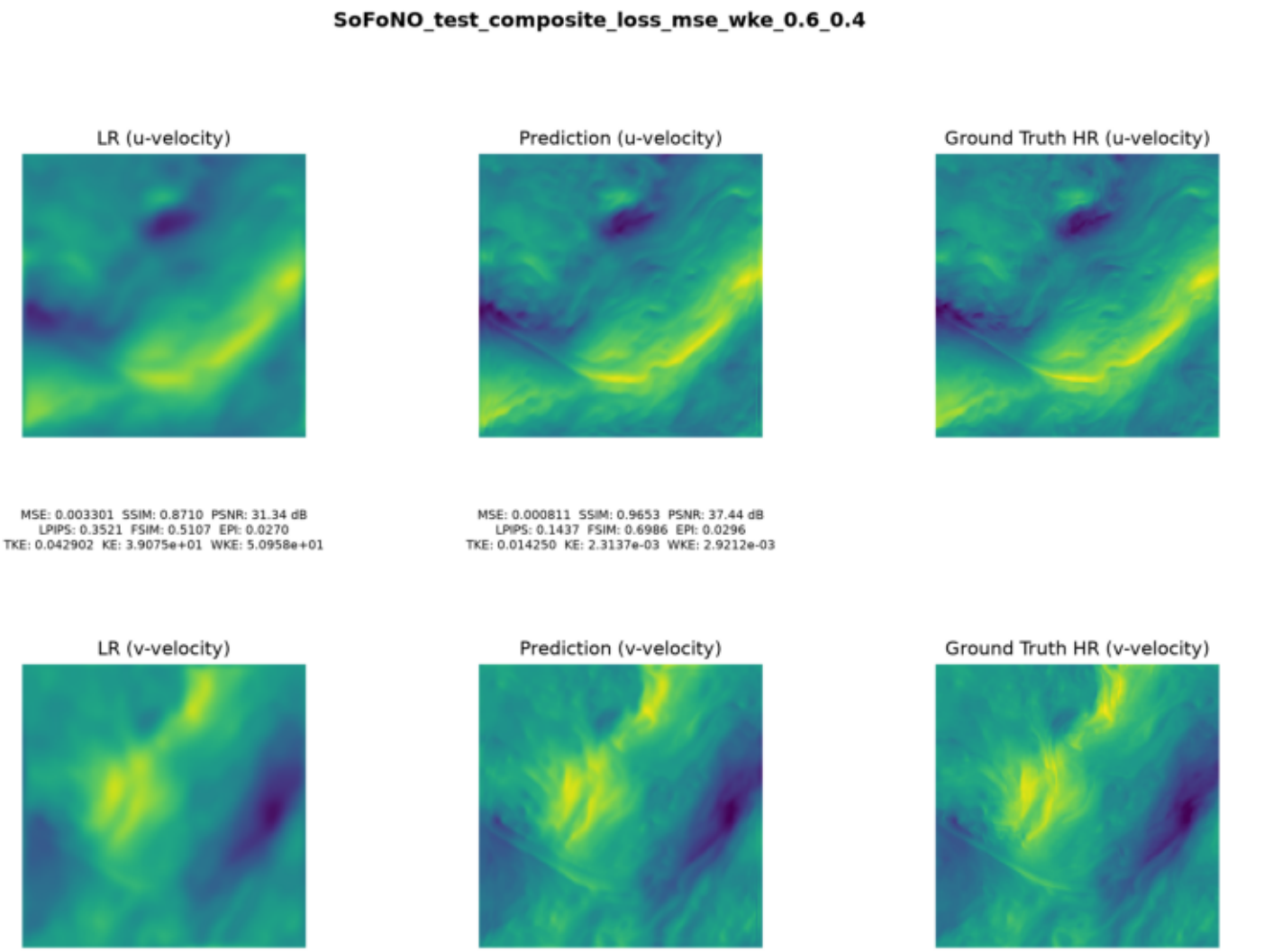




Top: kinetic-energy spectrum. Bottom: LR / Prediction / HR comparison. Both use the same fixed test-sample.

# SoFoNO_test_composite_loss_mse_wke_0.6_0.4

0.6 MSE + 0.4 Weighted KE




Top: kinetic-energy spectrum. Bottom: LR / Prediction / HR comparison. Both use the same fixed test-sample.

# SRNO -- Summary metrics (bold = best per column)

## u-velocity metrics

| Strategy | MSE | SSIM | PSNR | LPIPS | FSIM | EPI | TKE |
|---|---|---|---|---|---|---|---|
| MSE only | 0.0006147 | 0.9714 | 40.51 | 0.1614 | 0.7025 | **0.03144** | 0.01247 |
| 0.6 MSE + 0.4 LPIPS | **0.000145** | **0.9951** | 46.73 | **0.01422** | **0.9078** | 0.03002 | **0.001046** |
| 0.6 MSE + 0.2 LPIPS + 0.2 KE | 0.0001455 | 0.9943 | **46.8** | 0.01678 | 0.8989 | 0.03003 | 0.001291 |
| 0.6 MSE + 0.4 Weighted KE | 0.0009577 | 0.9712 | 38.62 | 0.1465 | 0.698 | 0.0296 | 0.01703 |

## v-velocity metrics

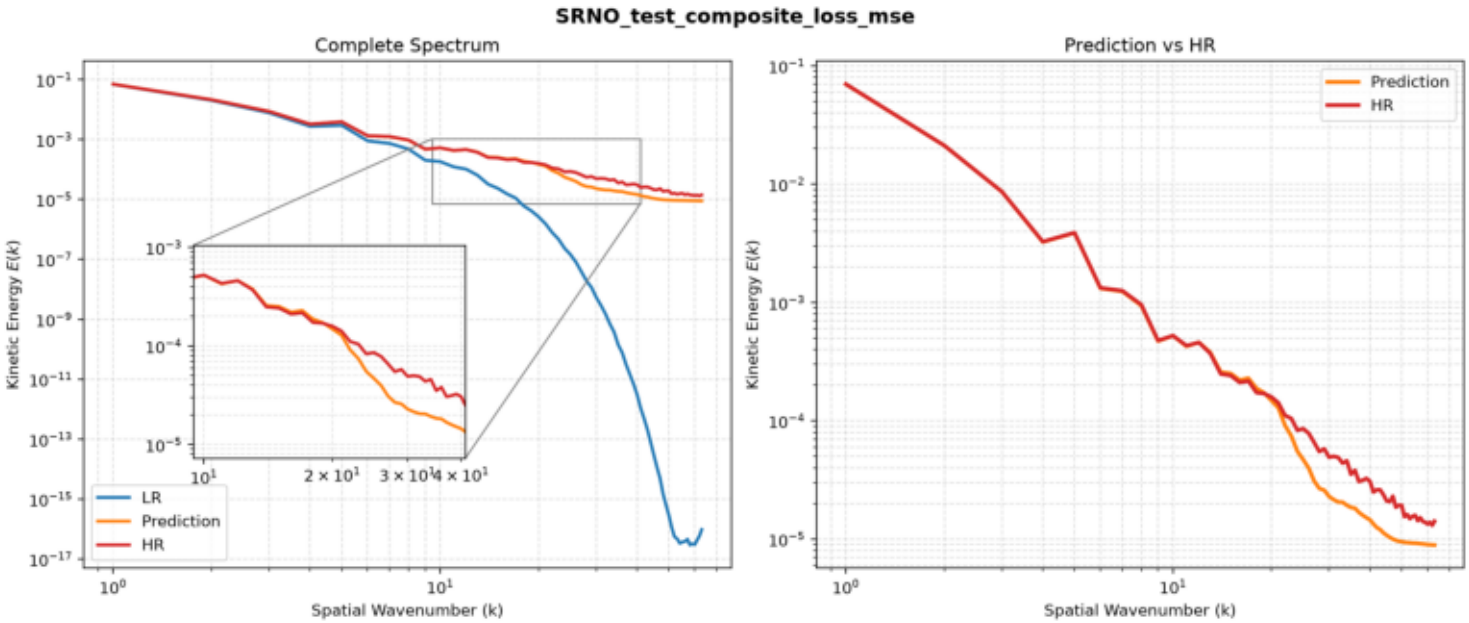
| Strategy | MSE | SSIM | PSNR | LPIPS | FSIM | EPI | TKE |
|---|---|---|---|---|---|---|---|
| MSE only | 0.0006332 | 0.9747 | 41.21 | 0.1532 | 0.7738 | **0.02801** | 0.01288 |
| 0.6 MSE + 0.4 LPIPS | **0.0001538** | **0.9955** | **47.31** | **0.01392** | **0.9319** | 0.02658 | **0.001114** |
| 0.6 MSE + 0.2 LPIPS + 0.2 KE | 0.0001686 | 0.9947 | 46.99 | 0.01643 | 0.9234 | 0.02656 | 0.001427 |
| 0.6 MSE + 0.4 Weighted KE | 0.0008523 | 0.9728 | 39.96 | 0.1263 | 0.7657 | 0.02485 | 0.01563 |

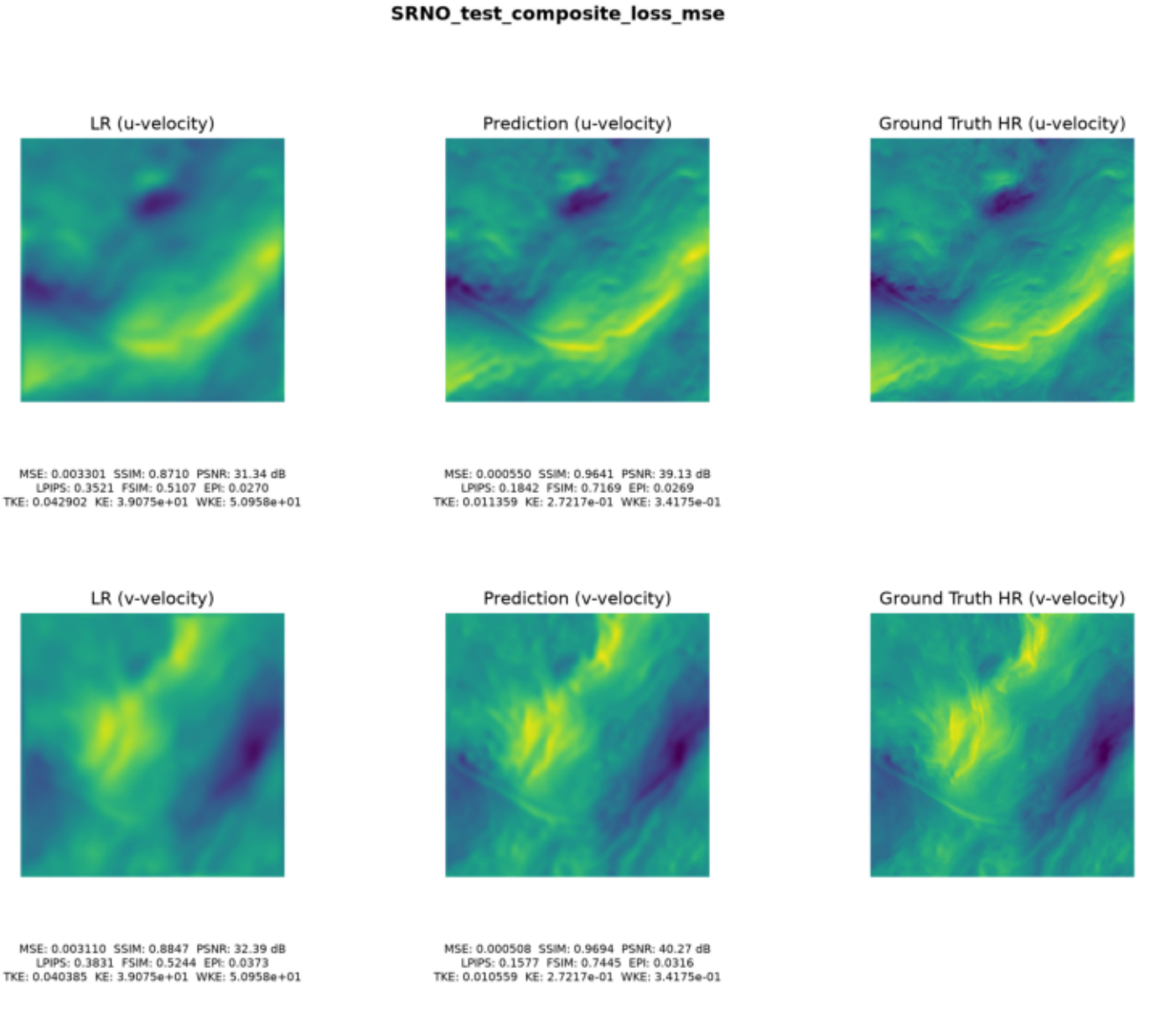
## Combined metrics (KE / WKE mix u and v, not split by channel)

| Strategy | KE | WKE |
|---|---|---|
| MSE only | 0.1658 | 0.2086 |
| 0.6 MSE + 0.4 LPIPS | 0.009156 | 0.01538 |
| 0.6 MSE + 0.2 LPIPS + 0.2 KE | **0.001141** | **0.001788** |
| 0.6 MSE + 0.4 Weighted KE | 0.003199 | 0.004069 |

# SRNO_test_composite_loss_mse

MSE only

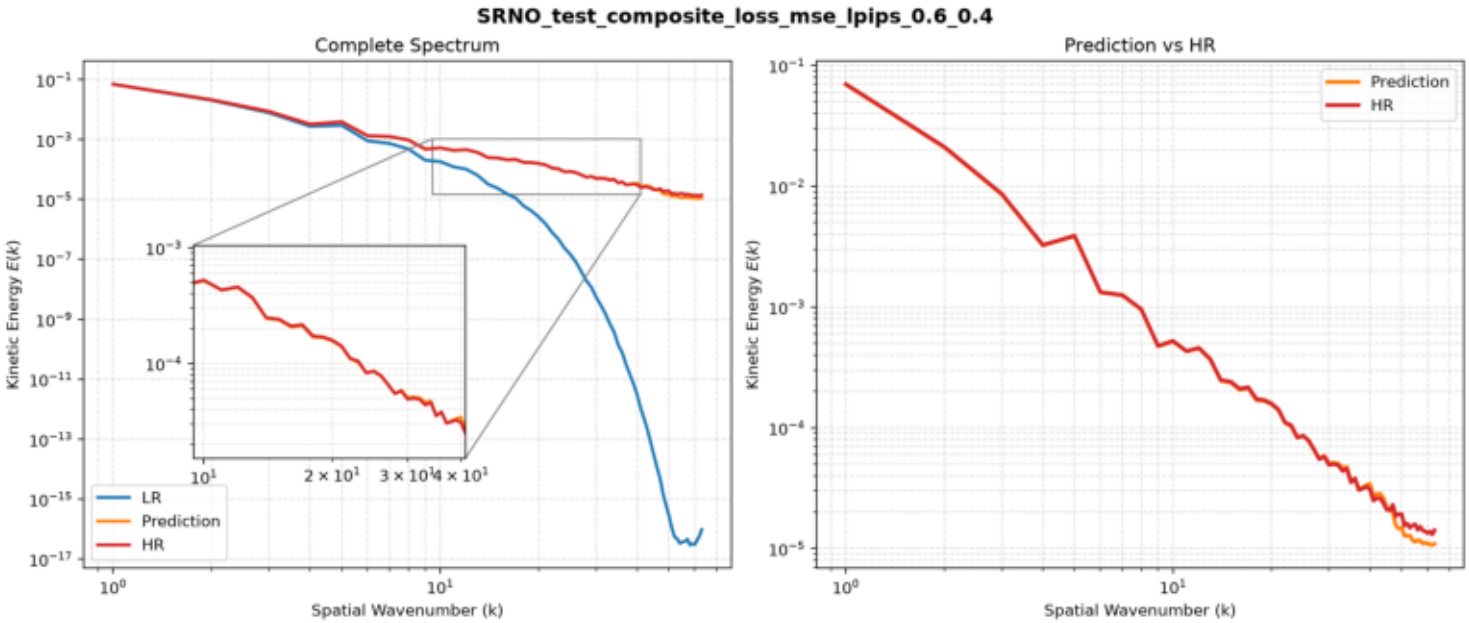


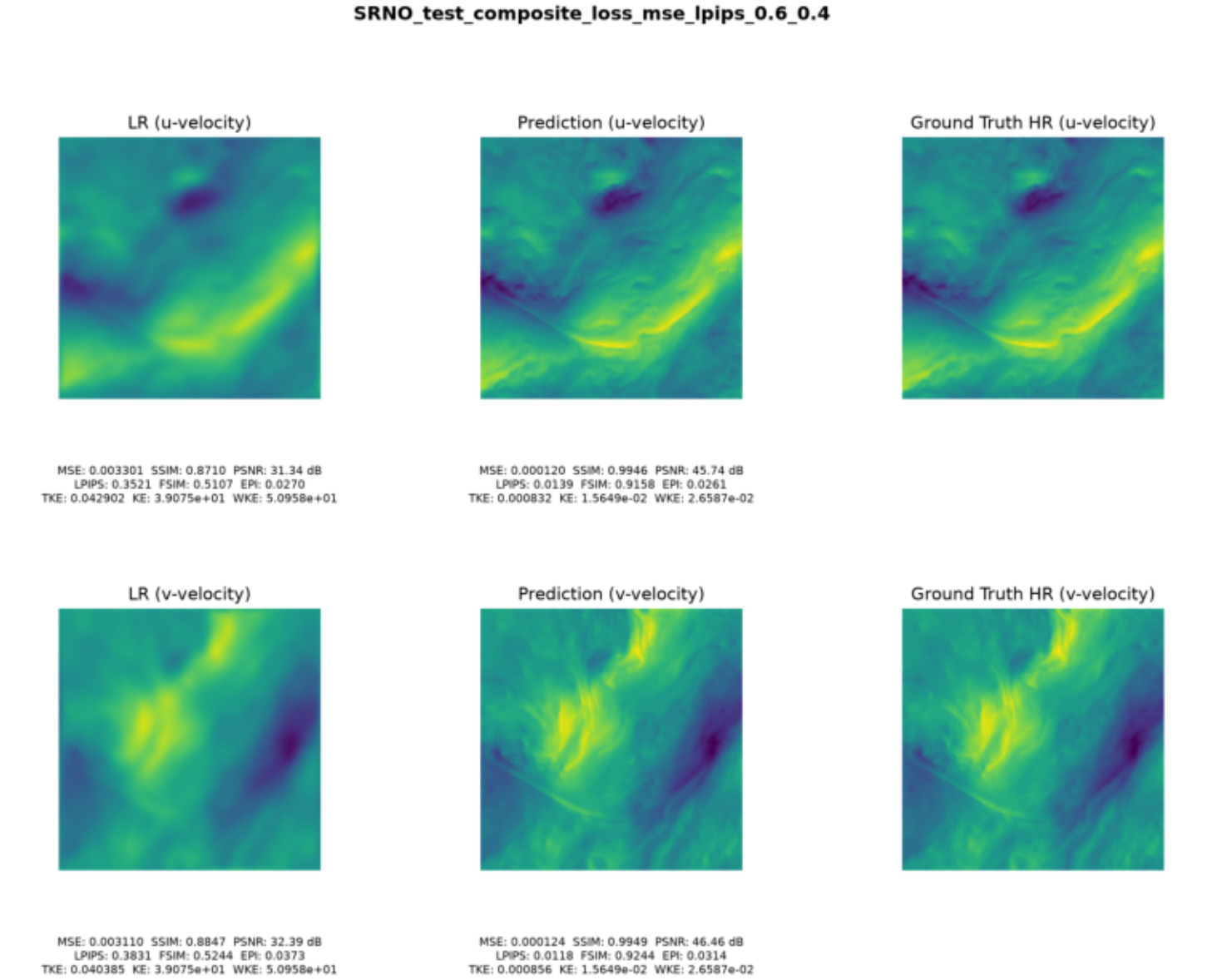




Top: kinetic-energy spectrum. Bottom: LR / Prediction / HR comparison. Both use the same fixed test-sample.

# SRNO_test_composite_loss_mse_lpips_0.6_0.4

0.6 MSE + 0.4 LPIPS

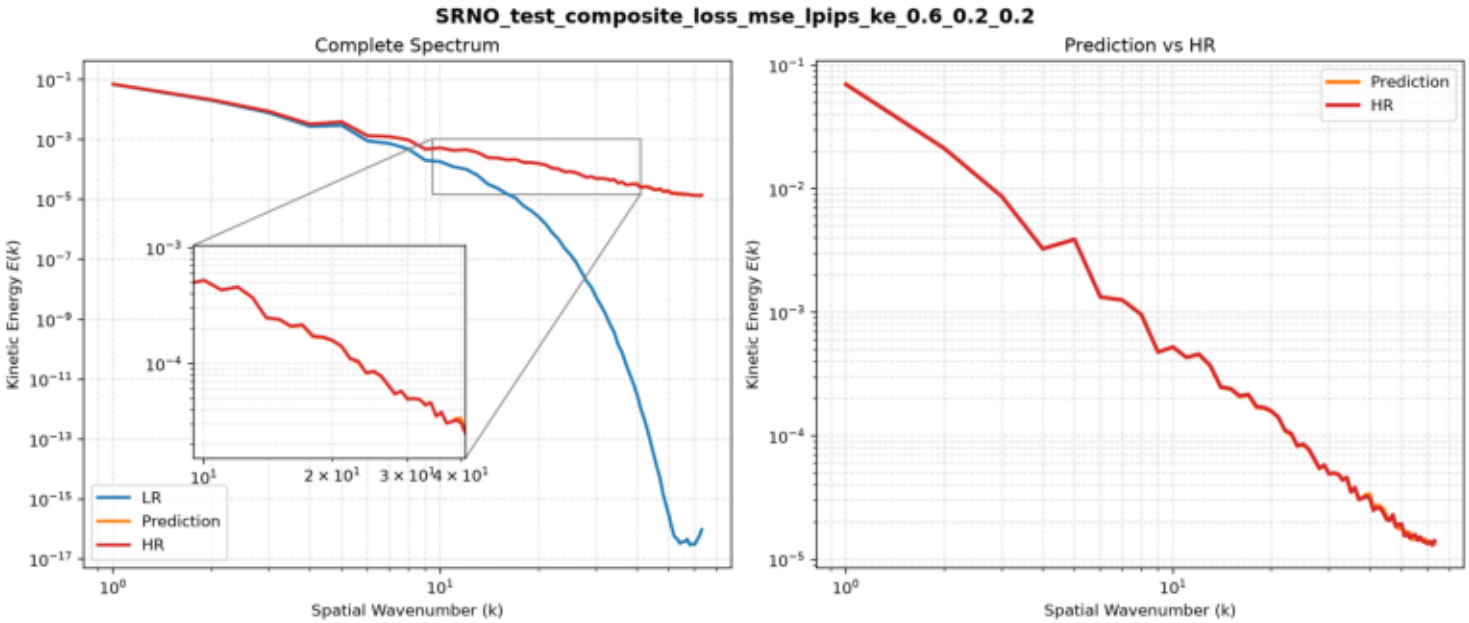


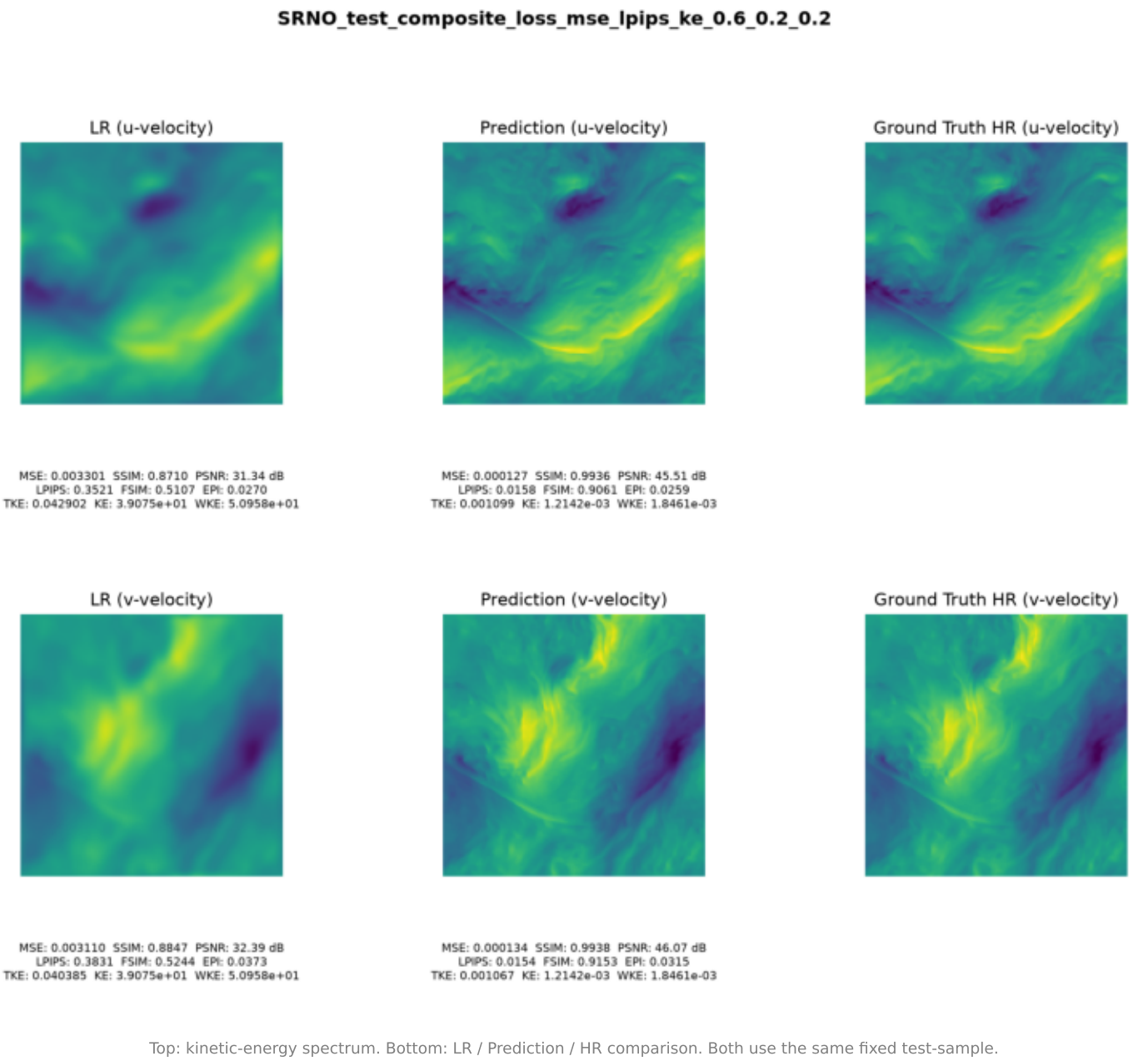




Top: kinetic-energy spectrum. Bottom: LR / Prediction / HR comparison. Both use the same fixed test-sample.

# SRNO_test_composite_loss_mse_lpips_ke_0.6_0.2_0.2

0.6 MSE + 0.2 LPIPS + 0.2 KE

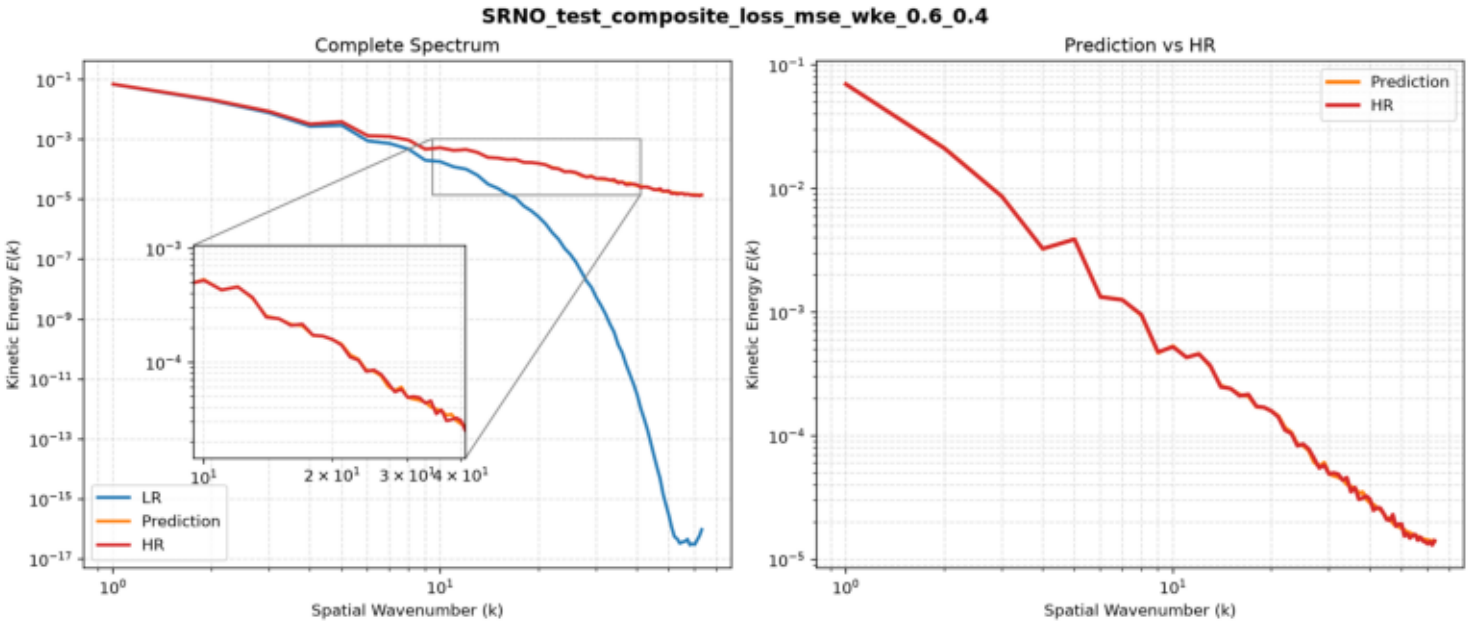


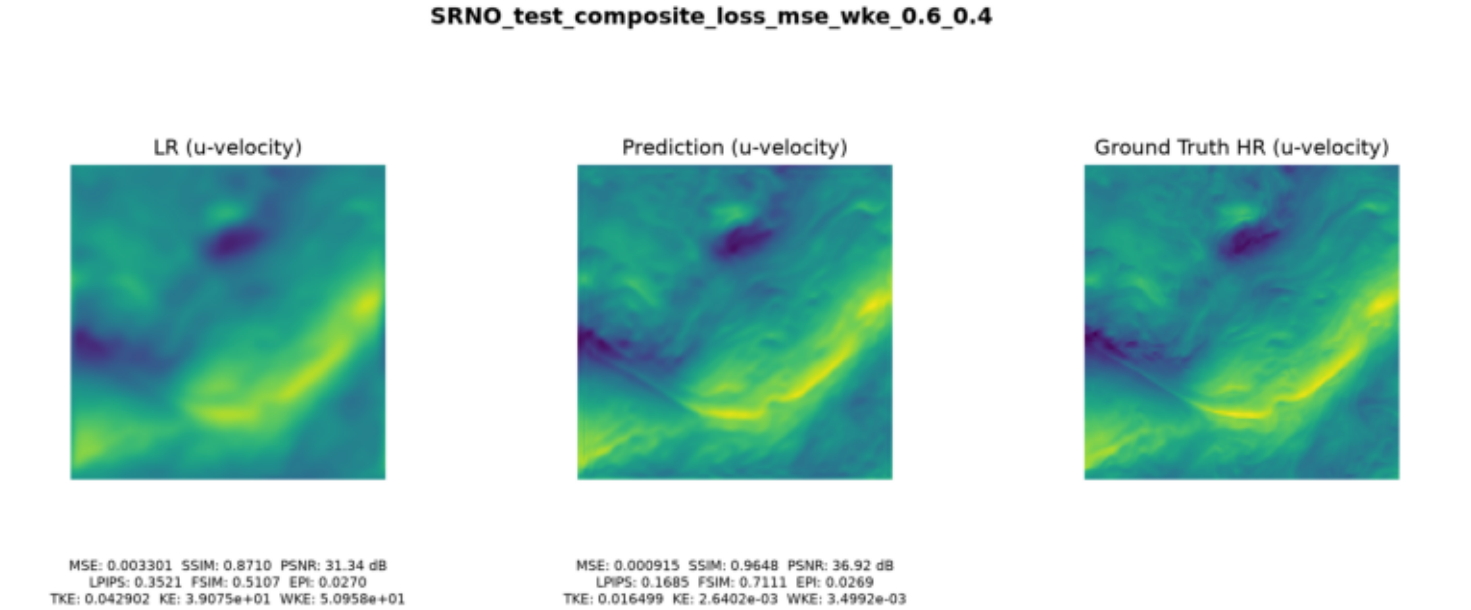



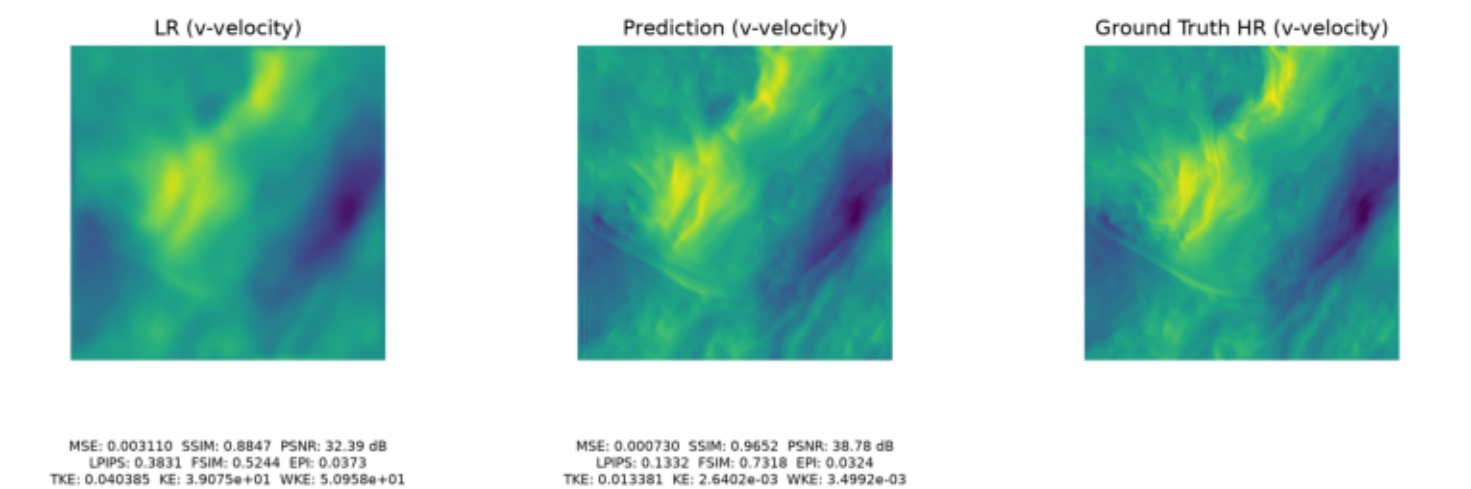

Top: kinetic-energy spectrum. Bottom: LR / Prediction / HR comparison. Both use the same fixed test-sample.

# SRNO_test_composite_loss_mse_wke_0.6_0.4

0.6 MSE + 0.4 Weighted KE




Top: kinetic-energy spectrum. Bottom: LR / Prediction / HR comparison. Both use the same fixed test-sample.